

1999-2000

वार्षिक प्रतिवेदन Annual Report



Human Resource Development for Efficiency, Effectiveness, and Excellence



राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

राजेन्द्रनगर, हैदराबाद-500 030, भारत

National Academy of Agricultural Research Management

Rajendranagar, Hyderabad - 500 030, India

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Foreword

NAARM Annual Report 1999-2000 reflects the work done by the NAARM faculty. This report was compiled and finalized under the leadership of Dr S.N. Saha, who was the Acting Director during my deputation to Centre for Development Research (ZEF), University of Bonn, Bonn, Germany. Since I have contributed little because of my absence from NAARM, the whole credit for and responsibility of this Report rests with Dr S.N. Saha, the then Acting Director.

Hyderabad
October 2000

J.C. Katyal
(J.C. Katyal)
Director



Preface

The year 1999-2000 has been very active and satisfying to a great extent since faculty members of NAARM feel happy that they have been able to contribute towards human resource development of NARS. As a participant in the process of bringing about change in attitude, value system, and commitment in agricultural scientists, NAARM has played a significant role by conducting training programmes, seminars, workshops, and interactive sessions. Brainstorming sessions addressing policy and function-level problems, with the participation of administrative and finance functionaries and scientists of NARS, brought out important suggestions for solving the difficult impediments that are crippling the system. The partnership role of NAARM has been visible in the process of education management and teacher development, in which TNAU and ANGRAU have benefited most by way of enhanced performance of their teachers and other employees, and have been adjudged as best performing universities in the country. The international collaboration in research and training programmes with ISNAR, Tanzania, and ICRA, The Netherlands, provided excellent experience, which will show a positive impact on our performance in the long time to come. For the purpose of enhancing the capability of NAARM faculty, we have been able to train some of our faculty members in some US universities and in Bangkok, with the financial support from AHRD project. Utilization of funds for maintaining and creating infrastructure facilities for the effective conduct of training programmes has been accomplished satisfactorily.

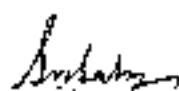
I wish to take this opportunity to put on record our sincere thanks and gratitude for the support, guidance, and encouragement received from Padma Bhusan Dr R.S. Paroda, Director General, ICAR, and Secretary, DARE, Govt. of India. NAARM is indebted to Dr S.L. Mehta, DDG (Education), Dr H.S. Nainawatee, ADG (HRD), and Mr K.L. Bokolia, Deputy Secretary (Education), for their care and nurturing attitude for the institutional growth and development of the Academy.

We, most humbly, record our sincere thanks to Dr J.C. Katyal, Director, NAARM, for sparing his valuable time for going through the draft, giving suggestions for improvement, and for empowering us to take all the steps in bringing out this annual report.

Dr P. Manikandan, Principal Scientist, with the able support from Shri R.V.V.S. Prakasa Rao, Editor, took all the trouble and the whole responsibility to collect the information and prepare the report in the present form. I sincerely express my thanks to them for their unstinted sincere co-operation. Dr A. Gopalam, Nodal Officer, AHRD Project, has compiled the theme

paper, which is timely, and it gives a detailed account of the project and the initiatives taken under this project for developing the Infrastructure as well as the faculty members of NAARM. I sincerely thank him for his efforts. All the faculty members and the Heads deserve appreciation for their timely supply of information. Mrs G. Aneesa's contribution in editing the theme paper and the help received from Mr K.V. Kumar in preparing the graphs are duly acknowledged. I also thank Mr P. Namdev and Mr L. Venkateswarlu for their help.

Hyderabad
October 2000


(S.N. Saha)

अप्रैल

- * कंप्यूटर जागरूकता पर एक कार्यक्रम 12-17 अप्रैल, 1999 तक आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के बारह वैज्ञानिक व तकनीकी अधिकारियों ने इसमें भाग लिया। डॉ.के.एम.रेड्डी इस कार्यक्रम के समन्वयक थे।
- * शैक्षिक वीडियो निर्माण पर दो सप्ताह का कार्यक्रम 12-24 अप्रैल, 1999 तक आयोजित किया गया। श्री के.आर. प्रभाकर द्वारा समन्वित इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के छः वैज्ञानिक और तकनीकी अधिकारियों ने भाग लिया।
- * कृषि अनुसंधान और शिक्षा में कंप्यूटर उपयोग पर कार्यपालक विकास कार्यक्रम, तमिलनाडु कृषि विश्वविद्यालय के वरिष्ठ अधिकारियों हेतु 14-19 अप्रैल, 1999 तक आयोजित किया गया। इसमें टी.एन.ए.यू. के पंद्रह वरिष्ठ अधिकारियों ने भाग लिया। इस कार्यक्रम का समन्वय डॉ.के.विद्यासागर राव और डॉ.आर.चित्तिरैचेत्वन ने किया।
- * एल.ए.एन, नेटवर्किंग और एम.एस.ऑफिस पर 21-23 अप्रैल, 1999 तक तीन दिवसीय कार्यशाला का आयोजन अकादमी में भा.कृ.अनु.प. द्वारा किया गया। भा.कृ.अनु.प. संस्थानों के बारह सदस्यों ने इसमें भाग लिया। इस कार्यक्रम का समन्वय एच.सी.एल. द्वारा किया गया।
- * एन.ए.टी.पी. की समीक्षा बैठक 25-26 अप्रैल, 1999 तक आयोजित की गई। इसमें एन.ए.टी.पी. के उत्पादन पद्धति अनुसंधान तत्व के अन्तर्गत अनुसंधान कार्यक्रमों में हुई प्रगति की समीक्षा की गई। डॉ.टी.बालगुरु द्वारा समन्वित इस कार्यक्रम में तैत्तलीस प्रतिनिधियों ने भाग लिया।

मई

- * शैक्षिक मनोविज्ञान और आचरण प्रौद्योगिकी पर टी.एन.ए.यू. के संकाय सदस्यों के लिए 10-14 मई, 1999 तक एक संकाय विकास कार्यक्रम आयोजित किया गया। इसमें टी.एन.ए.यू. के सत्रह वरिष्ठ आचार्यों ने भाग लिया। इसका समन्वय डॉ.ए.गोपालम और डॉ.आर.कल्पना शास्त्री ने किया।

जून

- * डाटा विश्लेषण हेतु सांख्यिकीय साफ्टवेयर पर एक प्रशिक्षण कार्यक्रम 01-11 जून, 1999 तक आयोजित किया गया। भा.कृ.अनु.प. संस्थानों, रा.कृ.वि. और अन्य संगठनों के दस वैज्ञानिक और तकनीकी अधिकारियों को इससे लाभ हुआ। इसका समन्वय डॉ.के.विद्यासागर राव ने किया।

* आंध्र प्रदेश के पशुपालन अधिकारियों हेतु उन्नत विस्तार और प्रशिक्षण पद्धति पर 02-12 जून, 1999 तक एक प्रायोजित कार्यक्रम अक्कादमी में आयोजित किया गया। इस विभाग के बीस अधिकारियों ने इस कार्यक्रम भाग लिया। डॉ.बीई.सवारत्नम और डॉ.एस.के.नन्दा इसके समन्वयक थे।

* कृषि अनुसंधान और विस्तार हेतु शोध, विश्रान्त, सहभागिता ग्रामीण मूल्यांकन तकनीक पर 22 जून से 02 जुलाई, 1999 तक एक पाठ्यक्रम का आयोजन किया गया। इसमें बारह सहभागियों ने भाग लिया, जिनमें भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के वैज्ञानिक, कृषि विज्ञान केंद्र और कृषक प्रशिक्षण केंद्रों के प्रशिक्षक शामिल थे। इसका समन्वय डॉ.बीई.सवारत्नम ने किया।

* आचरण प्रौद्योगिकी और शैक्षिक मनोविज्ञान पर 26 जून से 10 जुलाई, 1999 तक दो सप्ताह का संकाय विकास कार्यक्रम आयोजित किया गया। टी.एन.ए.यू.के पंद्रह संकाय सदस्यों ने इसमें भाग लिया। डॉ.गोपालम, डॉ.पी.मणिक्जुडन और डॉ.एन.हनुमंत राव ने इस कार्यक्रम का समन्वय किया।

जुलाई

* श्रम प्रबंध को गत्यात्मकता पर एक कार्यशाला 15-18 जुलाई, 1999 तक आयोजित की गई। इस कार्यशाला को इंस्टीट्यूट ऑफ पब्लिक इंटरप्राइज (आई.पी.ई) हैदराबाद के सहयोग से आयोजित किया गया। डॉ.आर.कल्पना शास्त्री और श्री बी.पाण्डु रेड्डी ने कार्यक्रम का समन्वयन किया। इसमें छब्बीस प्रतिनिधियों ने भाग लिया।

* कृषि अनुसंधान और शिक्षा में कंप्यूटर उपयोग पर कार्यपालक विकास कार्यक्रम टी.एन.ए.यू. के वरिष्ठ संकाय सदस्यों के लिए 19-24 जुलाई, 1999 तक आयोजित किया गया। इस कार्यक्रम द्वारा पंद्रह संकाय सदस्यों को लाभ हुआ। डॉ.एम.नारायण रेड्डी, डॉ.अजीत पी. मारु और डॉ.ए.गोपालम इसके समन्वयक थे।

* कृषि अनुसंधान मूल्यांकन पद्धति पर एक प्रशिक्षण कार्यक्रम 26-31 जुलाई, 1999 तक आयोजित किया गया। डॉ.के.पी.सी.राव द्वारा समन्वित इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के चौदह कृषि अर्थशास्त्री और सांख्यिकीविदों ने भाग लिया।

* कृषि अनुसंधान एवं शिक्षा में कंप्यूटर उपयोग पर 26-31 जुलाई, 1999 तक टी.एन.ए.यू. के वरिष्ठ संकाय सदस्यों हेतु कार्यपालक विकास कार्यक्रम आयोजित किया गया। तमिलनाडु कृषि विश्वविद्यालय के सोलह संकाय सदस्यों ने इस कार्यक्रम में भाग लिया। डॉ.डी.रामाराव और डॉ.आर.कल्पना शास्त्री ने इसका समन्वयन किया।

* कृषि में कंप्यूटर उपयोग पर दो सप्ताह का पाठ्यक्रम 27 जुलाई से 07 अगस्त, 1999 तक आयोजित किया

गया। एन.ए.आर.एस. के विभिन्न संस्थानों के सात वैज्ञानिक और तकनीकी अधिकारियों ने इसमें भाग लिया। डॉ.के.विद्यासागर राव और डॉ.के.एम.रेड्डी ने इसका समन्वयन किया।

अगस्त

* पुस्तकालय और सूचना केंद्रों में कंप्यूटर उपयोग पर एक पाठ्यक्रम 17-28 अगस्त, 1999 तक आयोजित किया गया। डॉ.के.विद्यासागर राव द्वारा समन्वित इस कार्यक्रम में एन.ए.आर.एस. के विभिन्न संस्थानों के तेरह पुस्तकालय और सूचना केंद्र के प्रबंध के लिए जिम्मेदार वैज्ञानिक और अधिकारियों ने भाग लिया।

* कृषि शिक्षा प्रबंध पर तीन दिवसीय कार्यशाला 19-21 अगस्त, 1999 तक आयोजन किया गया। इसमें कृषि विश्वविद्यालयों के बत्तीस संकायाध्यक्षों ने भाग लिया। इसका समन्वय डॉ.ए.गोपालम और डॉ.एस.एन.साहा ने किया।

* शैक्षिक मनोविज्ञान और आचरण प्रौद्योगिकी पर 23 अगस्त-06 सितम्बर, 1999 तक टी.एन.ए.यू. के वरिष्ठ संकाय सदस्यों हेतु संकाय विकास कार्यक्रम आयोजित किया गया। इस कार्यक्रम द्वारा इक्कीस संकाय सदस्यों को लाभ हुआ। डॉ.ए.गोपालम, डॉ.पी.कुमार और डॉ.आर.घितरैचेल्वन ने इस प्रायोजित कार्यक्रम का समन्वयन किया।

अक्तूबर

* उन्नत डाटा विश्लेषण हेतु सांख्यिकीय साफ्टवेयर पर दस दिवसीय पाठ्यक्रम 05-16 अक्तूबर, 1999 तक आयोजित किया गया। भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के सात वरिष्ठ स्तर के वैज्ञानिकों ने इस कार्यक्रम में भाग लिया। डॉ.के.विद्यासागर राव और डॉ.एम.नारायण रेड्डी ने कार्यक्रम का समन्वयन किया।

* कृषि में निर्णय समर्थन पद्धति पर 11-21 अक्तूबर, 1999 तक एक लघु पाठ्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के इक्कीस वैज्ञानिक और संकाय सदस्यों ने भाग लिया। डॉ.डी.रामाराव और डॉ.एस.के.नन्दा इस कार्यक्रम के समन्वयक थे।

* कृषि अनुसंधान का दूसरा प्रबंध विकास कार्यक्रम 25-30 अक्तूबर, 1999 तक आयोजित किया गया। इसमें चौदह विभागाध्यक्ष और परियोजना समन्वयकों ने भाग लिया। डॉ.एस.एन.साहा और डॉ.जगन्नाथम चेल्ला ने इसका समन्वयन किया।

* कृषि अनुसंधान और शिक्षा में कंप्यूटर उपयोग पर 25-30 अक्तूबर, 1999 तक टी.एन.ए.यू. के वरिष्ठ संकाय सदस्यों के लिए कार्यपालक विकास कार्यक्रम आयोजित किया गया। डॉ.एम.नारायण रेड्डी और डॉ.एन.हनुमंत राव द्वारा समन्वित इस कार्यक्रम में पंद्रह संकाय सदस्यों ने भाग लिया।

नवम्बर

* स्थाई कृषि विकास के लिए उत्पादन पद्धति उपागम पर 16-20 नवम्बर, 1999 तक एक पाठ्यक्रम आयोजित किया गया। एन.ए.आर.एस. के विभिन्न संगठनों के छः अनुसंधानकर्ताओं ने इसमें भाग लिया। डॉ.टी.बालगुरु और डॉ.पी.मणिकण्डन इसके समन्वयक थे।

* मानव संसाधन प्रबंध पर एक उन्नत पाठ्यक्रम आंध्र प्रदेश पशुचिकित्सा विभाग के अधिकारियों हेतु 22-27 नवम्बर, 1999 तक आयोजित किया गया। इसमें दस सहायक निदेशकों ने भाग लिया। डॉ.पी.मणिकण्डन और डॉ.आर.कल्पना शास्त्री इस आयोजित कार्यक्रम के समन्वयक थे।

* राजभाषा कार्यान्वयन के पचास वर्ष : उपलब्धियाँ और आकांक्षाएँ पर 29-30 नवम्बर, 1999 को दो दिवसीय कार्यशाला का आयोजन किया गया। इसमें भा.कृ.अनु.प. के विभिन्न संस्थानों में हिन्दी कार्य से जुड़े सत्तावन सहभागियों ने भाग लिया।

* शैक्षिक वीडियो उत्पादन पर 29 नवम्बर से 10 दिसम्बर, 1999 तक एक कार्यक्रम आयोजित किया गया। विभिन्न भा.कृ.अनु.प. संस्थानों तथा रा.कृ.वि. के आठ वैज्ञानिक और तकनीकी अधिकारियों ने इसमें भाग लिया। श्री के.आर.प्रभाकर ने इसका समन्वयन किया।

* भा.कृ.अनु.प. संस्थानों में प्रशासनिक एवं वित्तीय प्रबंध सुधार पर 30 नवम्बर से 06 दिसम्बर, 1999 तक एक पुनश्चर्चा पाठ्यक्रम आयोजित किया गया। इसमें विभिन्न भा.कृ.अनु.प. संस्थानों के वतीस वित्त एवं लेखा अधिकारियों ने भाग लिया। डॉ.एस.एन.साहा और श्री बी.पाण्डु रेड्डी इसके समन्वयक थे।

दिसम्बर

* एम.एस.ऑफिस और ई-मेल पर त्रिलहन अनुसंधान निदेशालय के कर्मचारियों हेतु 06-10 दिसम्बर, 1999 तक पाँच दिवसीय प्रशिक्षण कार्यक्रम का आयोजन किया गया। इस कार्यक्रम द्वारा बारह कर्मचारियों को लाभ हुआ। इसका समन्वय डॉ.अजित पी. मारु ने किया।

* कृषि वृद्धि और बौद्धिक सम्पदा अधिकार पर 07-09 दिसम्बर, 1999 तक इंस्टीट्यूट ऑफ पब्लिक इंटरप्राइज, हैदराबाद के सहयोग से एक तीन दिवसीय कार्यशाला आयोजित की गई। इस कार्यक्रम में पच्चीस वैज्ञानिक और निर्णय निर्माताओं ने भाग लिया। डॉ.आर.कल्पना शास्त्री और डॉ.के.पी.सी.राव इस कार्यशाला के समन्वयक थे।

- * कृषि में कंप्यूटर उपयोग पर 13-24 दिसम्बर, 1999 तक एक कार्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के सत्रह वैज्ञानिक और तकनीकी अधिकारियों ने इसमें भाग लिया। डॉ.के.एम.रेड्डी और डॉ.के.विद्यासागर राव इसके समन्वयक थे।
- * मानव संसाधन प्रबंध पर एक उन्नत पाठ्यक्रम, आंध्र प्रदेश पशुपालन विभाग के अधिकारियों हेतु 13-18 दिसम्बर, 1999 तक आयोजित किया गया। इसमें विभाग के अठारह सहायक निदेशकों ने भाग लिया। इस प्रायोजित कार्यक्रम के समन्वयक डॉ.पी.मणिकण्डन और डॉ.एस.शण्मुगम थे।
- * शिक्षा शास्त्र और शिक्षण प्रौद्योगिकी पर 13-14 दिसम्बर, 1999 तक दो दिवसीय कार्यशाला लैरी ई. मिल्लर, ओहियो स्टेट विश्वविद्यालय द्वारा ए.एच.आर.डी. के तहत आयोजित की गई। विभिन्न राज्य कृषि विश्वविद्यालय के सत्राईस संकाय सदस्यों ने इस कार्यशाला में भाग लिया। डॉ.बी.एस.चन्देल इस कार्यशाला के समन्वयक थे।
- * शैक्षिक मनोविज्ञान और आचरण प्रौद्योगिकी पर 20 दिसम्बर, 1999 से 03 जनवरी, 2000 तक टी.एन.ए.यू. कोयंबतूर के संकाय सदस्यों के लिए संकाय विकास कार्यक्रम का आयोजन किया गया। इसमें टी.एन.ए.यू. के विभिन्न कॉलेजों के इकतीस वरिष्ठ स्तर के संकाय सदस्यों ने भाग लिया। डॉ.ए.गोपालम, डॉ.एन.हनुमंत राव और डॉ.के.पी.सी. राव इसके समन्वयक थे।
- * कृषि अनुसंधान प्रबंध पर 21-23 दिसम्बर, 1999 तक एक कार्यपालक विकास कार्यक्रम आयोजित किया गया। इसमें नौ सहभागी जिनमें भा.कृ.अनु.प. विभिन्न संस्थानों के निदेशक और विभागाध्यक्ष भी शामिल थे। डॉ.एस.एन.साहा और डॉ.जगन्नाथम चल्ला इसके समन्वयक थे।

जनवरी, 2000

- * चंद्रशेखर आज़ाद कृषि और प्रौद्योगिकी विश्वविद्यालय, कानपुर के वरिष्ठ स्तर के आचार्यों हेतु कृषि अनुसंधान प्रबंध का प्रायोजित कार्यक्रम 10-15 जनवरी, 2000 तक आयोजित किया गया। इसमें पच्चीस वरिष्ठ आचार्य और विभागाध्यक्षों ने इसमें भाग लिया। डॉ.टी.बालगुरु और डॉ.पी.मणिकण्डन इस कार्यक्रम के समन्वयक थे।
- * इंटरनेट और इंटरनेट प्रशासन पर एक कार्यक्रम 17-23 जनवरी, 2000 तक आयोजित किया गया। तमिलनाडु कृषि विश्वविद्यालय के बारह संकाय सदस्यों ने इस कार्यक्रम में भाग लिया। डॉ.अजित पी. मारु इस कार्यक्रम के समन्वयक रहे।
- * राजभाषा नीति पर पाँच दिवसीय कार्यशाला व प्रशिक्षण 18-22 जनवरी, 2000 के दौरान आयोजित किया

गया। डॉ.ए.गोपालम एवं श्रीमती जे.रेणुका ने इस कार्यक्रम का समन्वयन किया, जिसमें चौत्तीस कार्याधिकारियों ने भाग लिया।

* तिलहन नीति संबंधी मर्दों पर 22 जनवरी, 2000 के दौरान एक दिवसीय कार्यशाला आयोजित की गई। इस कार्यक्रम में पच्चीस वैज्ञानिकों ने भाग लिया। डॉ.डी.रामा राव एवं डॉ.आर.कल्पना शास्त्री, इस कार्यशाला के समन्वयक रहे।

* 24-31 जनवरी, 2000 के दौरान इंटरनेट के लिए विषय सामग्री विकास पर पाठ्यक्रम आयोजित किया गया। डॉ.अजित पी. मारू द्वारा समन्वयन किए हुए इस प्रायोजित कार्यक्रम में तमिलनाडु कृषि विश्वविद्यालय के बारह संकाय सदस्यों ने भाग लिया।

* तमिलनाडु कृषि विश्वविद्यालय के प्रशासनिक एवं वित्तीय कर्मचारियों के लिए कृषि अनुसंधान एवं शिक्षा में कंप्यूटर प्रयोग पर दि. 24-30 जनवरी, 2000 के दौरान एक कार्यकारी विकास कार्यक्रम आयोजित किया गया। डॉ.के.विद्यासागर राव, डॉ.एम.नारायण रेड्डी एवं डॉ.ए.गोपालम इस कार्यक्रम का समन्वयन किया, जिसमें पंद्रह सहभागियों ने भाग लिया।

* 24-30 जनवरी, 2000 के दौरान तमिलनाडु कृषि विश्वविद्यालय के प्रशासनिक एवं वित्तीय कर्मचारियों के लिए प्रौद्योगिक प्रशासन एवं वित्तीय मर्दों के प्रबंध पर कार्यकारी विकास कार्यक्रम आयोजित किया गया। इस कार्यक्रम से पच्चीस सहभागी लाभान्वित हुए। डॉ.के.पी.सी.राव, डॉ.एन.संध्या शेनार्थ एवं डॉ.ए.गोपालम इस कार्यक्रम के समन्वयक रहे।

फरवरी, 2000

सूचना केंद्रों के प्रबंध पर 07-18 फरवरी, 2000 के दौरान एक पाठ्यक्रम आयोजित किया गया। इस पाठ्यक्रम के समन्वयक डॉ.अजित पी.मारू रहे, जिसमें सात ऑरिस प्रकोष्ठ के प्रभारी तथा नेटवर्क प्रशासकों ने भाग लिया।

* भा.कृ.अनु.प. के प्रशासनिक एवं वित्तीय अधिकारियों के लिए संगठनात्मक एवं प्रबंध सुधार पर दि 13-15 फरवरी, 2000 के दौरान एक प्रथम प्रतिक्रियात्मक कार्यशाला आयोजित की गई। भा.कृ.अनु.प. संस्थानों के चौरासी प्रशासनिक एवं वित्त तथा लेखा अधिकारियों ने इस कार्यशाला में भाग लिया। डॉ. जगन्नाथम घल्ला ने इस कार्यशाला का समन्वयन किया।

* 16-17 फरवरी, 2000 के दौरान सुग्राह्यता तथा अभिमुखीकरण द्वारा संस्थागत प्रबंध के सुधार हेतु दो दिवसीय मस्तिक मंथन सत्र आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के चौरासी प्रशासनिक तथा वित्त एवं लेखा अधिकारियों ने इस सत्र में भाग लिया। डॉ.के.पी.सी.राव एवं डॉ.एन.हनुमंत राव इस कार्यक्रम के समन्वयक रहे।

- * तमिलनाडु कृषि विश्वविद्यालय के संकाय सदस्यों के लिए विशेषतः शैक्षणिक मनोविज्ञान एवं आचरण प्रौद्योगिकी पर 21-26 फरवरी, 2000 के दौरान एक संकाय विकास कार्यक्रम आयोजित किया गया। इस कार्यक्रम में सत्रह संकाय सदस्यों ने भाग लिया। डॉ.के.पी.सी.राव, डॉ.एन.हनुमंत राव एवं डॉ.ए.गोपालम इस कार्यक्रम के समन्वयक रहे।
- * फरवरी 21-26, 2000 के दौरान माइक्रोसाफ्ट परियोजना के प्रयोग द्वारा परियोजना प्रबंध पर एक कार्यक्रम आयोजित किया गया। इस कार्यक्रम में एन.ए.टी.पी. के कुल तेईस प्रधान अन्वेषकों ने भाग लिया। डॉ.डी.रामा राव और डॉ.अजित पी. मारु ने इस कार्यक्रम का समन्वयन किया।
- * 'प्रशिक्षकों का प्रशिक्षण' पर पाठ्यक्रम 22 फरवरी से 04 मार्च, 2000 तक आयोजित किया गया। इस कार्यक्रम में विभिन्न संस्थानों के पाँच प्रशिक्षण आयोजकों ने भाग लिया, जिसके समन्वयक डॉ.टी.बालगुरु एवं डॉ.पी.मणिकण्डन रहे।
- * तमिलनाडु कृषि विश्वविद्यालय के प्रशासनिक एवं वित्तीय कर्मचारियों हेतु दि 28 फरवरी से 04 मार्च, 2000 के दौरान कृषि अनुसंधान और शिक्षा में कंप्यूटर प्रयोग पर एक कार्यकारी विकास कार्यक्रम आयोजित किया गया। डॉ.एन.नारायण रेड्डी, डॉ.अजित पी. मारु और डॉ.आर.चित्तरैचेल्वन ने इस कार्यक्रम का समन्वयन किया, जिसमें पंद्रह सहभागियों ने भाग लिया।
- * 28 फरवरी से 09 मार्च, 2000 के दौरान तमिलनाडु कृषि विश्वविद्यालय के प्रशासनिक एवं वित्तीय कर्मचारियों के लिए शैक्षणिक प्रशासन एवं वित्तीय प्रबंध पर एक कार्यकारी विकास कार्यक्रम आयोजित किया गया। इस कार्यक्रम के समन्वयक डॉ.के.पी.सी.राव, डॉ.ए.गोपालम एवं डॉ.पी.मणिकण्डन रहे, जिससे तेईस सहभागी लाभान्वित हुए।

मार्च, 2000

- * तमिलनाडु कृषि विश्वविद्यालय के वरिष्ठ स्तर के संकाय सदस्यों व अधिकारियों के लिए 06-13 मार्च, 2000 के दौरान इंटरनेट और इंटरनेट प्रशासन पर एक कार्यक्रम आयोजित किया गया। डॉ.अजित पी.मारु इस कार्यक्रम के समन्वयक रहे, इसमें अठारह सहभागियों ने भाग लिया।
- * तमिलनाडु कृषि विश्वविद्यालय के वरिष्ठ स्तर के संकाय सदस्यों व अधिकारियों के लिए 14-21 मार्च, 2000 के दौरान इंटरनेट के लिए विषय सामग्री विकास पर एक कार्यक्रम आयोजित किया गया। इस कार्यक्रम के समन्वयक डॉ.अजित पी.मारु रहे जिससे अठारह सहभागी लाभान्वित हुए।
- * 27 मार्च - 01 अप्रैल, 2000 के दौरान कृषि में इंटरनेट प्रयोग पर एक प्रशिक्षण कार्यक्रम आयोजित किया गया। कृषि अनुसंधान एवं विकास संस्थानों में कार्यरत कुल दस वैज्ञानिक एवं अधिकारियों ने इस कार्यक्रम में भाग लिया। डॉ.आर.चित्तरैचेल्वन एवं डॉ.अजित पी. मारु ने इस कार्यक्रम का समन्वयन किया।

* कृषि अनुसंधान व शिक्षा में कंप्यूटर प्रयोग पर एक कार्यकारी विकास कार्यक्रम 27 मार्च से 01 अप्रैल, 2000 के दौरान आयोजित किया गया। इस कार्यक्रम में तमिलनाडु कृषि विश्वविद्यालय से तेरह प्रशासनिक एवं वित्तीय कर्मचारियों ने भाग लिया। डॉ.एम.नारायण रेड्डी, डॉ.एन.हनुमंत राव व डॉ.के.एम.रेड्डी इस कार्यक्रम के समन्वयक रहे।

* 27 मार्च - 01 अप्रैल, 2000 के दौरान तमिलनाडु कृषि विश्वविद्यालय के अधिकारियों के लिए व्यक्तिगत रूप से शिक्षा प्रशासन एवं वित्तीय मुद्दों के प्रबंध पर कार्यकारी विकास कार्यक्रम आयोजित किया गया। तमिलनाडु कृषि विश्वविद्यालय के तेईस प्रशासनिक एवं वित्तीय कर्मचारियों ने इस कार्यक्रम में भाग लिया। डॉ.के.पी.सी.राव, श्री बी.पाण्डु रेड्डी एवं डॉ.ए.गोपालम ने इस कार्यक्रम का समन्वयन किया।

Activities at a Glance

April

- * A programme on Computer Awareness was organized from April 12 to 17, 1999. Eleven scientists and technical officers of various ICAR Institutes participated in the programme. Dr K.M. Reddy co-ordinated the programme.
- * A two-week course on Educational Video Production was organized from April 12 to 24, 1999. Six scientists and technical officers of various ICAR institutes participated in the programme, co-ordinated by Mr K.R. Prabhakar.
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized for the senior officers of Tamil Nadu Agricultural University (TNAU), from April 14 to 19, 1999. Fifteen senior officers of TNAU participated in the programme. Dr K. Vidyasagar Rao and Dr R. Chithiraichelvan co-ordinated the programme.
- * A three-day course on LAN, Networking, and MS Office was organized at the Academy, by ICAR, from April 21 to 23, 1999. Twelve members from ICAR institutes participated in the programme. The programme was co-ordinated by HCL.
- * NATP Review Meeting was organized on April 25 and 26, 1999, to review the progress of research programmes under the production system research component of NATP. The programme, co-ordinated by Dr T. Balaguru, was attended by 43 delegates.

May

- * A Faculty Development Programme on Educational Psychology and Behavioural Technology was organized for the faculty members of TNAU, from May 10 to 14, 1999. Seventeen senior-level Professors from TNAU participated in the programme. Dr A. Gopalam and Dr R. Kalpana Sastry co-ordinated the programme.

June

- * A training programme on Statistical Software for Data Analysis was organized from June 1 to 11, 1999. Ten scientists and technical officers from ICAR Institutes, SAUs, and other organizations were benefited by the programme, co-ordinated by Dr K. Vidyasagar Rao.
- * A sponsored programme on Advanced Extension and Training Methodology for Animal Husbandry Personnel of Andhra Pradesh was organized from June 2 to 12, 1999. Twenty officers of the Department attended the programme. Dr VE. Sabarathnam and Dr S.K. Nanda co-ordinated the programme.
- * A course on Rapid, Relaxed, and Participatory Rural Appraisal Techniques for Research and Extension in Agriculture was organized from June 22 to July 2, 1999. Scientists from ICAR Institutes and SAUs, trainers from Krishi Vigyan Kendras and Farmers' Training Centres, numbering 12, participated in the programme, co-ordinated by Dr VE. Sabarathnam.

- * A two-week Faculty Development Programme on Behavioural Technology and Educational Psychology was organized from June 26 to July 10, 1999. Twenty five faculty members from TNAU participated in the programme. Dr A. Gopalam, Dr P. Manikandan, and Dr N. Hanumantha Rao co-ordinated the programme.

July

- * A workshop on Dynamics of Labour Management was organized from July 15 to 18, 1999. The workshop was organized in collaboration with the Institute of Public Enterprise (IPE), Hyderabad. Dr R. Kalpana Sastry and Mr B. Pandu Reddy co-ordinated the programme, which was attended by 26 delegates.
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized for the senior faculty members of TNAU, from July 19 to 24, 1999. Fifteen faculty members were benefited by this programme. Dr M. Narayana Reddy, Dr Ajit P. Maru, and Dr A. Gopalam co-ordinated this programme.
- * A training programme on Agricultural Research Evaluation Methods was organized from July 26 to 31, 1999. Fourteen Agricultural Economists and Agricultural Statisticians from ICAR Institutes and SAUs participated in this programme, co-ordinated by Dr K.P.C. Rao
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized for the senior faculty members of TNAU, from July 26 to 31, 1999. Sixteen faculty members of Tamil Nadu Agricultural University attended the programme. Dr D. Rama Rao and Dr R. Kalpana Sastry co-ordinated the programme.
- * A two-week course on Computer Applications in Agriculture was organized from July 27 to August 7, 1999. Seven scientists and technical officers from various institutions of NARS attended the programme. Dr K. Vidyasagar Rao and Dr K.M. Reddy co-ordinated the programme.

August

- * A course on Computer Applications in Libraries and Information Centres was organized from August 17 to 28, 1999. Thirteen scientists and officers of different Institutions under NARS, who are responsible for the management of libraries and information centres, participated in the programme, co-ordinated by Dr K. Vidyasagar Rao.
- * A three-day workshop on Management of Agricultural Education was organized from August 19 to 21, 1999. Thirty two Deans from various Agricultural Universities participated in the workshop. This workshop was co-ordinated by Dr A. Gopalam and Dr S.N. Saha.
- * A Faculty Development Programme on Educational Psychology and Behavioural Technology was organized for the senior faculty members of TNAU, from August 23 to September 6, 1999. Twenty one faculty members were benefited by participation in this programme. Dr A. Gopalam, Dr P. Kumar, and Dr R. Chithiraichelvan co-ordinated this sponsored programme.

October

- * A ten-day course on Statistical Softwares for Advanced Data Analysis was organized from October 5 to 16, 1999. Seven senior-level scientists from ICAR institutes and SAUs participated in the programme. Dr K. Vidyasagar Rao and Dr M. Narayana Reddy co-ordinated the programme.
- * A short course on Decision Support Systems in Agriculture was organized from October 11 to 21, 1999. Twenty one scientists and faculty members from various ICAR institutes and SAUs participated in the programme. Dr D. Rama Rao and Dr S.K. Nanda were the co-ordinators of this programme.
- * The 2nd Management Development Programme in Agricultural Research was organized from October 25 to 30, 1999. Fourteen Heads of the Departments and Project Co-ordinators participated in the programme. Dr S.N. Saha and Dr Jagannadham Challa co-ordinated this programme.
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized for the senior faculty members of TNAU, from October 25 to 30, 1999. Fifteen faculty members participated in this programme, co-ordinated by Dr M. Narayana Reddy and Dr N. Hanumantha Rao.

November

- * A course on Production Systems Approach for Sustainable Agricultural Development was organized from November 16 to 20, 1999. Researchers from different organizations of NARS, numbering six, participated in the programme. Dr T. Balaguru and Dr P. Manikandan co-ordinated this programme.
- * An Advanced Course on Management of Human Resources was specifically organized for the officers of the Department of Animal Husbandry, Andhra Pradesh, from November 22 to 27, 1999. Ten Assistant Directors participated in this course. Dr P. Manikandan and Dr R. Kalpana Sastry were the co-ordinators of this sponsored programme.
- * A two-day workshop on 50 years of Official Language Implementation: Achievements and Expectations was organized on November 29 and 30, 1999. A total of 57 participants, from various ICAR institutes, associated with Hindi activities in their institutes, took part in the workshop.
- * A programme on Educational Video Production was organized from November 29 to December 10, 1999. Eight scientists and technical officers from various ICAR institutes and SAUs participated in the programme, co-ordinated by Mr K.R. Prabhakar.
- * A refresher course on Improving Administrative and Financial Management in ICAR Institutes was organized from November 30 to December 6, 1999. Thirty two Administrative

Officers and Finance and Accounts Officers of various ICAR institutes attended this programme. Dr S.N. Saha and Mr B. Pandu Reddy co-ordinated the programme.

December

- * A five-day training on MS Office and E-mail was organized for the staff members of Directorate of Oilseeds Research, Hyderabad, from December 6 to 10, 1999. Twelve staff members were benefited through this programme. Dr Ajit P. Maru co-ordinated the programme.
- * A three-day workshop on Intellectual Property Rights and Agricultural Growth was organized in collaboration with the Institute of Public Enterprise (IPE), Hyderabad, from December 7 to 9, 1999. The programme had the participation of 25 scientists and decision-makers. Dr R. Kalpana Sastry and Dr K.P.C. Rao were the co-ordinators of this workshop.
- * A programme on Computer Applications in Agriculture was organized from December 13 to 24, 1999. Seventeen scientists and technical officers from ICAR institutes and SAUs attended the programme. Dr K.M. Reddy and Dr K. Vidyasagar Rao co-ordinated the programme.
- * An Advanced Course on Management of Human Resources was specifically planned and organized for the officers of the Department of Animal Husbandry, Andhra Pradesh, from December 13 to 18, 1999. Eighteen Assistant Directors of the Department participated in this sponsored programme. Dr P. Manikandan and Dr S. Shanmugam co-ordinated this programme.
- * A two-day workshop on Pedagogy and Educational Technology was organized, at the Academy, by Lary E. Miller of Ohio State University, under the aegis of AHRD, on December 13 and 14, 1999. Twenty seven faculty members from various SAUs participated in the workshop. Dr B.S. Chandel was the co-ordinator of this workshop.
- * A Faculty Development Programme on Educational Psychology and Behavioural Technology was organized from December 20, 1999, to January 3, 2000, for the faculty members of TNAU, Coimbatore. Thirty one senior-level faculty members from different colleges of TNAU participated in the programme. Dr A. Goplam, Dr N. Hanumantha Rao, and Dr K.P.C. Rao co-ordinated this programme.
- * An Executive Development Programme on Agricultural Research Management was organized from December 21 to 23, 1999. Nine Heads of the Departments and Directors of various ICAR institutes participated in the programme. Dr S.N. Saha and Dr Jagannadham Challa co-ordinated the programme.

January 2000

- * A sponsored programme on Agricultural Research Management was organized from January 10 to 15, 2000, for the senior-level Professors of Chandrasekhar Azad University of Agriculture and Technology, Kanpur. Twenty five senior Professors and Heads of the Departments participated in this course. Dr. T. Balaguru and Dr P. Manikandan were the co-ordinators of this programme.

- * A programme on Internet and Intranet Administration was organized from January 17 to 23, 2000. Twelve faculty members from Tamil Nadu Agricultural University participated in the programme. Dr Ajit P. Maru was the co-ordinator of the programme.
- * A five-day Workshop-cum-Training on Official Language Policy was organized from January 18 to 22, 2000. Dr A. Gopalam and Ms J. Renuka co-ordinated the programme, which was attended by 34 officials.
- * A one-day workshop on Oilseeds Policy Issues was organized on January 22, 2000. Twentyfive scientists participated in the programme. Dr D. Rama Rao and Dr R. Kalpana Sastry co-ordinated the workshop.
- * A course on Content Development for Intranets was organized from January 24 to 31, 2000. Twelve faculty members of Tamil Nadu Agricultural University participated in this sponsored programme, co-ordinated by Dr Ajit P. Maru.
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized for the administrative and finance personnel of Tamil Nadu Agricultural University, from January 24 to 30, 2000. Dr K. Vidyasagar Rao, Dr M. Narayana Reddy, and Dr A. Gopalam co-ordinated this programme, which was attended by 15 participants.
- * An Executive Development Programme on Educational Administration and Management of Financial Issues was organized for the administrative and finance personnel of Tamil Nadu Agricultural University, from January 24 to 30, 2000. Twenty five participants were benefited by attending this programme, which was co-ordinated by Dr K.P.C. Rao, Dr N. Sandhya Shenoy, and Dr A. Gopalam.

February 2000

- * A course on Management of Information Centres was organized from February 7 to 18, 2000. The course, co-ordinated by Dr Ajit P. Maru, was attended by seven ARIS Cell in-charges and network administrators.
- * First Interactive Workshop on Organizational and Management Reforms for Administrative and Finance Officers of ICAR was organized from February 13 to 15, 2000. Eighty four Administrative and Finance and Accounts Officers of ICAR institutes participated in the workshop. Dr Jagannadham Challa co-ordinated this workshop.
- * A two-day Brainstorming Session on Betterment of Institutional Management Through Sensitization and Orientation was organized on February 16 and 17, 2000. Eighty four Administrative and Finance and Accounts Officers of ICAR Institutes participated in the session. Dr K.P.C. Rao and Dr N. Hanumantha Rao co-ordinated this programme.
- * A Faculty Development Programme on Educational Psychology and Behavioural Technology was organized exclusively for the faculty members of Tamil Nadu Agricultural

University, from February 21 to 26, 2000. Seventeen faculty members attended the programme. Dr K.P.C. Rao, Dr N. Hanumantha Rao, and A Gopalam co-ordinated the programme.

- * A programme on Project Management Using Microsoft Project was organized from February 21 to 26, 2000. Twenty three Principal Investigators of NATP participated in the programme. Dr D. Rama Rao and Dr Ajit P. Maru co-ordinated the programme.
- * A course on Training of Trainers was organized from February 22 to March 4, 2000. Five training organizers from various institutions participated in the programme, co-ordinated by Dr T. Balaguru and Dr P. Manikandan.
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized from February 28 to March 4, 2000, for the administrative and finance personnel of Tamil Nadu Agricultural University. Dr M. Narayana Reddy, Dr Ajit P. Maru, and Dr R. Chithiraichelvan co-ordinated this programme, attended by 15 participants.
- * An Executive Development Programme on Educational Administration and Management of Financial Issues was organized for the administrative and finance personnel of Tamil Nadu Agricultural University, from February 28 to March 9, 2000. The programme, co-ordinated by Dr K.P.C. Rao, Dr A. Gopalam, and Dr P. Manikandan, benefited 23 participants.

March 2000

- * A programme on Internet and Intranet Administration was organized for the senior-level faculty members and officers of Tamil Nadu Agricultural University, from March 6 to 13, 2000. Dr Ajit P. Maru co-ordinated the programme, which was attended by 18 participants.
- * A programme on Content Development for Intranets was organized for the senior-level faculty members and officers of Tamil Nadu Agricultural University, from March 14 to 21, 2000. The programme, co-ordinated by Dr Ajit P. Maru, benefited 18 participants.
- * A training programme on Internet Usage in Agriculture was organized from March 27 to April 1, 2000. The programme had the participation of 10 scientists and officers working in agricultural R & D institutions. Dr R. Chithiraichelvan and Dr Ajit P. Maru co-ordinated the programme.
- * An Executive Development Programme on Computer Applications in Agricultural Research and Education was organized from March 27 to April 1, 2000. Thirteen administrative and finance personnel from TNAU attended the programme. Dr M. Narayana Reddy, Dr N. Hanumantha Rao, and Dr K.M.Reddy co-ordinated this programme.
- * An Executive Development Programme on Educational Administration and Management of Financial Issues was exclusively organized for the officials of TNAU, from March 27 to April 1, 2000. The programme was attended by 23 administrative and finance personnel of TNAU. Dr K.P.C. Rao, Mr B. Pandu Reddy, and Dr A. Gopalam co-ordinated the programme.

To support changes in policy and structure in the National Agricultural Research System (NARS) and to achieve higher levels of professional management in agriculture, the National Academy of Agricultural Research Management (NAARM) was established in 1976. NAARM is one of the five National Institutes of the Indian Council of Agricultural Research (ICAR), New Delhi, the apex body of the country for promoting agricultural research, education, and extension education.

Since its inception, the Academy has been taking up the responsibility of organizing training programmes, Executive Development Programmes, Faculty Development Programmes, workshops, and seminars in the areas related to agricultural research and education management for the scientists at different levels in the NARS and also for the scientists of the third world countries. Research on management problems associated with agricultural research and education is another important activity of the Academy. NAARM has also been functioning as a repository of ideas and information, both at the national and international levels, and acting as a clearing house for dissemination and utilization of such information and knowledge. The Academy is contributing significantly to generate a sense of fraternity and inculcate scientific culture among the agricultural scientists in the country. This unique institution has already established its credibility in the national and international arena. It is looked upon as a model institute by the developing countries and it is fast developing into a viable Regional Training Centre in this part of the world to train agricultural scientists in agricultural research and education management.

The Academy is located in the premises of Acharya N.G. Ranga Agricultural University in Rajendranagar. The Academy, spread over 50 hectares area, has enviable sylvan environs. Over the years, the Academy has been able to build up excellent infrastructure and facilities necessary for imparting quality training. The faculty members of the Academy are extensively trained across a range of areas covering agricultural research and education management, both in India and abroad. With the excellent facilities available for training and with the expertise and experience available with the faculty members, the Academy has been able to build its strength for taking up varied activities to fulfil its mandate.

Mandate

- ❖ To organize and conduct training programmes in agricultural research management for the scientists at various levels
- ❖ To build up high quality resource material in agricultural research management based on actual field experience
- ❖ To undertake systematic review and study of management problems of agricultural research institutes, programmes, and systems
- ❖ To plan, organize, and conduct workshops and seminars in research management and educational technology
- ❖ To organize, liaise, and co-ordinate programmes of international co-operation in the field of agricultural research management
- ❖ To assist the State Agricultural Universities (SAUs) in developing Regional Centres of Management to cater to the needs of various Universities

Objectives

- ❑ To organize foundation courses for the scientists newly recruited into the Agricultural Research Service of ICAR
- ❑ To design and organize training programmes, workshops, and seminars, and to develop high quality material in agricultural research and education management for the senior and middle-level scientists of NARS
- ❑ To play a major role in different processes of human resource development of NARS, in the context of agricultural research and education
- ❑ To undertake policy-level studies on prioritization of research and resource allocation, and to enhance research productivity of the constituents of NARS and their programmes
- ❑ To evolve suitable management techniques for effective planning, scheduling, organizing, monitoring, and evaluation of agricultural research and development projects in NARS
- ❑ To study the management issues of various programmes and patterns of technology transfer, develop effective systems and strategies for effective communication of agricultural technologies
- ❑ To undertake development programmes for evolving more effective and efficient

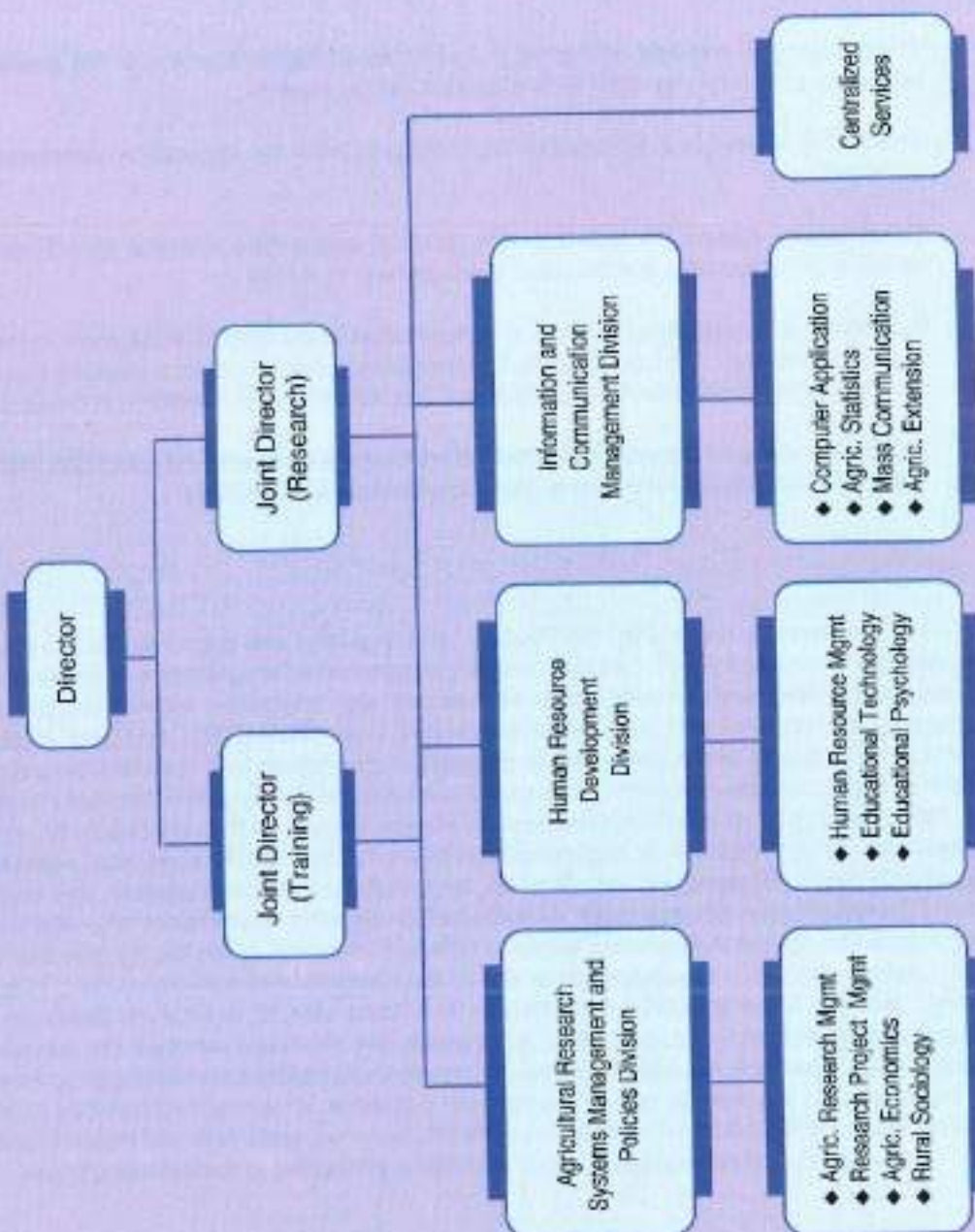
approaches, methods, and models in higher agricultural education for use in SAUs and Deemed Universities of ICAR

- ❑ To promote and execute academic programmes of higher learning in the specialized areas of agricultural research and education management
- ❑ To develop expertise in information technology to meet the application software needs of NARS
- ❑ To undertake systematic review and to develop appropriate systems for efficient and effective administrative and financial management in NARS
- ❑ To function as a repository of ideas and information in the form of databases in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas
- ❑ To offer consultancy services in the field of agricultural research and education management, and agricultural information and communication technology

Organizational Structure

The Academy is headed by the Director, who is guided and supported by the Institute Management Committee (IMC). The IMC reviews and promotes programmes of the Academy, and approves investments in new areas of research and education, training programmes, workshops, and seminars. The Research Advisory Committee (RAC) and Staff Research Council (SRC) provide broad guidelines and assist in developing and implementing specific research programmes and projects. The Director is assisted by two Joint Directors (currently one Joint Director is in position) and three Division Heads in the execution of various programmes. The Academy is organized into three functional Divisions, viz. Agricultural Research Systems Management and Policies, Human Resource Development, and Information and Communication Management. A multi-disciplinary team of scientists, who also constitute the core faculty of the Academy, works in different Divisions. Each faculty member is an agricultural scientist, with specialization in one of the disciplines of agriculture, who has also received advanced and specialized training in the broad area of agricultural research and education management. As needed and appropriate, the Academy borrows the services of outside resource persons, specialized in specific areas, for its multifarious training programmes. The functioning of the Academy is supported by a number of centralized services such as Administration and Accounts, Hostel Services, Health Services, and Press and Public Relations Cell. The current organizational structure of NAARM is presented in the following figure.

Organizational Structure of NAARM



Agricultural Human Resource Development Project - Implementation at the National Academy of Agricultural Research Management

Prologue

National Agricultural Research System in India, one of the largest in the world, with ICAR, SAUs, and state and central Governments as collaborating partners, is engaged in the development of agricultural research and education for the last 50 years. During the last decade or so, realization has come in that the quality of graduates, now produced by SAUs, has a mismatch with the anticipated developmental requirements of the society and community. The State Agricultural Universities and ICAR deemed-to-be Universities do not lead to skills that match with the requirements of the next millennium. Whereas there are many contributing factors, some of the most important factors include lack of opportunities for faculty competence improvement, poor infrastructure in educational institutions, extensive inbreeding, poor library facilities, lack of access to information technology, course curricula not in tune with global changes, and lack of proper environment for the proper grooming of the young scientists. To propel India into the 21st century, it is imperative that the agricultural human resource develop the skills and entrepreneurship that would enable them to be globally competitive. The Agricultural Human Resource Development Project (AHRDP), launched with the support from World Bank, in the soft loan mode, has the following objectives.

- To improve the quality relevance of higher agricultural education and in-service training
- To strengthen the capacity of the participating states to develop and manage agricultural human resource

With the above objectives, the AHRD project was launched on August 14, 1995, and will terminate on December 31, 2000. The participating agencies are Indian Council of Agricultural Research (ICAR), and the States of Andhra Pradesh, Haryana, and Tamil Nadu. The total cost of the project is US \$ 74.2 million, of which World Bank will provide US \$ 59.5 million, and the corresponding share of the Government of India and three participating states would have to be US \$14.7 million, as per the Standard Staff Appraisal Report specifications.

The Education Division of the Council has the responsibility to co-ordinate and monitor the implementation of the project, and the Deputy Director General (Education) is designated as the National Project Co-ordinator. The underlined statement of the project is to improve the quality of agricultural education by ensuring an all-round development of agriculture, with appropriate participation of developmental agencies.

AHRDP implementation at NAARM

Agricultural Human Resource Development Project, operating at NAARM, is focused at developing Educational Technology Cells, strengthening of existing facilities at the Academy, and conducting teacher improvement programmes for various agricultural universities. For the teacher improvement programme component, a number of training programmes were conducted for the faculty members of Tamil Nadu Agricultural University (TNAU), Coimbatore, Acharya N.G. Ranga Agricultural University, (ANGRAU), Hyderabad, and Tamil Nadu University of Veterinary and Animal Sciences (TANUVAS), Chennai, with the project funds assistance from these Universities, and at Rajasthan Agricultural University, Gujarat Agricultural University, and Punjab Agricultural University, by the funding provided by ICAR. Five senior-level workshops on Management of Agricultural Education were conducted for the Deans of State Agricultural Universities, for sensitizing them on the new leadership roles in the renewed professionalism settings.

a) Infrastructure Development

i) Upgradation of Video Instruction and Production Laboratory

Video being one of the powerful media of communication, the planned and efficient use of it in the training function was facilitated by upgrading the already available video instruction and production laboratory established under FAO-UNDP Project. Various equipment were procured with the project assistance which helped in improving the quality and effectiveness of video programmes produced and also in supporting various training programmes. Some of the equipment procured included Camcorders and VCRs in Betacam, Edit Controller, Computerized non-linear video editing system, U-matic SP and SVHS Video Cassette recorders, CD players, Speakers and headphones, LCD projectors, Computer with real time 3D studio max software, computer with Hindi font and video interface, Computers, and optical fibre networking. The video facilities are extended to the trainees of various programmes for developing the knowledge and skills. The scientist-trainees of the Foundation Course use the facilities to produce high quality video information shorts. Video is used in other programmes for providing feedback to the participants' performance in exercises, especially in behaviour-oriented situations. The video feedback enhances the effectiveness of instruction. Teachers of FDP are provided hands-on experience for developing video-based instructional segments, for integrating them in the classroom instruction.

ii) Optical fibre networking

In order to establish excellent communication within and outside NAARM, it was proposed to link the entire campus of NAARM by optical fibre network. Administrative Block, Academic Block, Faculty Block, and the non-academic areas were laid with optical fibre networking. E-mail, Internet, and Intranet facilities were provided, which are useful in establishing excellent communication. They also facilitate excellent search facilities for collecting relevant information which are required for training and research activities. All these have helped in improving the training effectiveness.

iii) Computers and LCD projectors

With the support of the project, computers, printers, scanners, LCD projectors, and various multi-utility hardware and software were procured. These, in turn, have facilitated upgradation of skills and competency enhancement of faculty members, in addition to enhancing the quality of training.

iv) Other infrastructure development

In addition to the procurement of various equipment for strengthening the training and other related activities, a Tata Sumo was also purchased under the project. The faculty seminar hall is proposed for modernization, in terms of furniture, acoustics, audio-visual projection, air-conditioning, etc.

b) Overseas Training of NAARM Faculty Members

The faculty members of the Academy need to be updated in their knowledge and skills in order to improve their capability and confidence. The project provided an opportunity for the development of the faculty members. Universities and institutions in USA, UK, and other countries, which provide training programmes related to agricultural education, and which are otherwise not available in India, are selected carefully for this purpose. The various places in which the faculty members underwent training included Ohio State University, Michigan State University, University of Illinois, and Cornell University from USA; Wye College, London; and Asian Institute of Technology, Bangkok. Nine faculty members were immensely benefited through this overseas training of three months duration. Various aspects related to agricultural education planning and management were the areas of training specialization. These faculty members brought back with them rich knowledge and skills, which are put to appropriate use in the various activities of the Academy.

c) Organization of Teacher Orientation Programmes

Under the project, various training programmes and workshops were organized for the State Agricultural Universities, which not only helped in improving the teacher effectiveness but also in crystallizing the ideas on management of agricultural education. Teacher orientation programmes were organized at Tamil Nadu University of Veterinary and Animal Sciences, Chennai; Rajasthan Agricultural University, Udaipur; Gujarat Agricultural University, Anand; and Punjab Agricultural University, Ludhiana. A total of 120 faculty members of these universities were immensely benefited by participation in these programmes. Four training programmes on Pedagogy were conducted by IBM Global, Bangalore, at NAARM, in which more than 50 faculty members were trained.

In order to improve the teaching skills and also to sensitize the teachers to the current educational needs, a total of 23 teacher improvement programmes were conducted, with the AHRDP funds of the participating SAUs, which benefited 570 faculty members of various levels.

These teachers were trained in educational methodology and instructional technology. Participants included faculty members from Tamil Nadu Agricultural University, Tamil Nadu University of Veterinary and Animal Sciences, and Acharya N.G. Ranga Agricultural University. These training programmes were conducted at two different levels - training for 17 days in the area of pedagogy to Associate and Assistant Professors, and an abridged training of 10 days on management related issues for senior faculty members. The feedback of these programmes strongly pointed out the need for such training programmes to facilitate development of teachers' qualities, both at instructional level as well as at behavioural level and also for improving the quality of agricultural education. In addition, training programmes on computer applications and on administrative and financial issues were organized for administrative and finance personnel. These programmes facilitated 157 personnel to develop the needed knowledge and skills.

d) Training Resource Material

Appropriate training resource materials were developed for various training programmes, organized under the project. Changes in the content of the programmes were brought about, based on the specific areas needed for interaction and also based on the structured and oral feedback received at the end of each of the programme. The cost of production of these training resource materials were met out of the training programme funds.

e) Deans Workshop on Management of Agricultural Education

Academic improvements in SAUs are spearheaded by the Deans and Associate Deans. To sensitize them to the management and decision-making skills, the Education Division of ICAR initiated workshops on "How to Manage Agricultural Education Better?" for the Deans and Associate Deans of SAUs. These workshops provided a forum for the Deans to discuss on the management problems of agricultural education and to identify strategies to take agricultural education into the 21st century, in the light of the recommendations of the M.S. Swaminathan Committee report on agricultural education. At the end of the workshops, an action plan was prepared by the participating Deans and Associate Deans, for their respective colleges. While doing so, the issues concerning the integration of gender equity were also given adequate consideration. These workshops were very timely and useful, and they had the active participation of 140 Deans and Associate Deans, drawn from various SAUs. The feedback indicated that these workshops provided the needed insight and information for the Deans to prepare meaningful curriculum for meeting the future demands of agricultural education.

Epilogue

Agricultural Human Resource Development Project, operating at NAARM, with the able support and assistance from the Education Division of ICAR, provided an opportunity for the development of overall functional capability of NAARM. The infrastructure, meant for training advancement, was developed with a view to supplement the earlier developments made through Plan allocations. Upgradation of the cost-intensive video instruction laboratory and the networking of communication facilities made NAARM globally acceptable, as evidenced by the

number of requests received from national and international organizations, universities, and institutions for collaboration with NAARM, in the areas of training and research. The exchange of ideas by cross-cultural interactions in the consultants-supported workshops and training programmes, provided an opportunity for projecting NAARM as a capable, vibrant, and dynamic organization. Extensive overseas training of the NAARM faculty members not only enhanced their own preparedness in the arising fields of educational technology as applied to agriculture, but also enabled them to develop newer areas for training and research. The project is at its terminal stage, realizing its listed objectives to near totality. There is a possibility of extension of its operation by one more year, to be used as a bridge time for battenning phase1 with Phase 2, being finalized now.

Training Programmes

Programme	Date	Activity
Foundation Courses		
Foundation Course for Agricultural Research Service		The foundation course for agricultural research service, which is a basic induction training for the scientists newly recruited to agricultural research service, with four months duration, is organized in three phases, Phase 1 : Agricultural research systems and management processes - concept and principles; Phase 2 : Technology generation and transfer process - field experience training; Phase 3 : Agricultural research systems and management processes - practice and applications. The foundation course aims at exposing the participants to the various aspects of agricultural research project management, in addition to developing adequate skills in them for managing their research, by inculcating inter-disciplinary team spirit. Adequate opportunities are provided to the scientists to develop knowledge and skills in information technology, communication, administration and finance, and in organizational behaviour. A feeling of fraternity is also developed and nurtured among the scientists during the period of the foundation training. The training also prepares them for the future demands of agricultural research, at the national as well as at the international level. During the period under report, three foundation courses were organized, with 114 scientists successfully completing the programme. Dr K.M. Reddy, Dr Janardan Jee, Dr P. Kumar, Dr N. Sandhya Shenoy, Dr R. Kalpana Sastry, and Dr S.K. Nanda were the co-ordinators of these FOCARS programmes.
FOCARS 67	01-06-99 28-09-99	
FOCARS 68	24-08-99 21-12-99	
FOCARS 69	06-01-2000 04-05-2000	
Foundation Course for Administrative Officers and Finance and Accounts Officers of ICAR Institutes	26-07-99 19-08-99	This course was intended for the directly recruited Administrative Officers and Finance and Accounts Officers of ICAR Institutes. The programme aimed at improving the effectiveness of the administrative officers as well as finance and accounts officers of the combined cadre in discharging their functions and in providing a constructive support to the Directors of the institutes in realizing the organizational goals and needs. Sixteen directly-recruited Administrative Officers and Finance and Accounts Officers of various ICAR institutes participated in the programme, co-ordinated by Dr S.N. Saha and Mr B. Pandu Reddy.

Senior Programmes

Computer 12-04-99
Awareness 17-04-99

In order to lend a helping hand to professionals in the agricultural sector who are striving to equip themselves with the basic awareness of the fundamentals and a working knowledge of computers, a one-week programme on Computer Awareness was organized. This training programme was exclusively designed to expose the participants from ICAR institutes, SAUs, and other agricultural organizations to the fundamentals of computers and to give an insight into the working of the most commonly used software in agricultural sector. Eleven scientists and technical officers from various ICAR institutes participated in this programme, co-ordinated by Dr K.M. Reddy.

Educational 12-04-99
Video Pro- 24-04-99
duction 29-11-99
 10-12-99

Educational video tapes are becoming increasingly important in the present day education system because of their graphic presentation and sound effects. Skills, technical mastery, and imagination are essential to produce educational video of high quality.

The Academy has established an Advanced Centre for professional video production and training. With the various facilities available, the Academy is in a very good position to provide hands-on experience to the participants for educational video production. The course aimed at equipping the participants with the necessary skills in the production of educational video programmes and familiarized them with the creative aspects of making educational videos. Fourteen scientists and technical officers of various ICAR institutes and SAUs were benefited through these two hands-on-intensive courses. The programmes were co-ordinated by Mr K.R. Prabhakar.

Statistical 01-06-99
Software for 11-06-99
Data Analy-
sis

With the advent of user-friendly software packages, analysis and interpretation of complex and large volume of data are no more a distant reality. In order to familiarize the participants with various advanced statistical software packages and to develop the needed confidence and skills in them to handle various packages, thereby enabling them to analyze the data and present the results more effectively, a course on Statistical Software for Data Analysis was organized. The course dealt in detail statistical software packages such as MSTATC and MICROSTAT, Statistical features of MS Excel, and also some graphic packages. Ten scientists and technical officers from

Rapid,
Relaxed,
and Participatory Rural
Appraisal
Techniques
for Research
and Extension in
Agriculture

22-06-99
02-07-99

Agricultural
Research
Evaluation
Methods

26-07-99
31-07-99

Computer
Applications
in Agriculture

27-07-99
07-08-99
13-12-99
24-12-99

various ICAR institutes, SAUs, and other organizations participated in this course. Dr K. Vidyasagar Rao was the co-ordinator of this programme.

The application of rapid, relaxed, and participatory rural appraisal techniques for the purpose of agricultural research and extension is gaining momentum now-a-days. The objective of this programme was to impart the required skills in the participants to use various appraisal techniques in the actual field situation and to enable them to link these techniques for project preparation in research and extension. The programme enabled the participants to learn the necessary knowledge and skills through concept awareness lectures, individual and group exercises, and field visits. The programme, co-ordinated by Dr VE. Sabarathnam, was attended by 12 scientists belonging to various institutions of NARS.

National Agricultural Research System is facing a widespread shortage of funds on one hand, while the accountability of the investments made is increasing on the other. In order to impress the policy makers to allocate more funds, it is necessary to carry out a widespread evaluation of agricultural research at project, programme, regional, and national levels. The tool kit of agricultural research evaluation is quite rich, varied, and interesting. Newer methods are also constantly evolved for research evaluation. It is important that the methodology as well as applications of agricultural research evaluation are carried out competently. A one-week programme on Agricultural Research Evaluation Methods was, therefore, organized at the Academy to train the agricultural economists and statisticians in agricultural research evaluation methods. Fourteen Agricultural Economists and Agricultural Statisticians, from different ICAR institutes and SAUs, participated in this programme, co-ordinated by Dr K.P.C. Rao.

The developments in computer science and information technology have a profound impact on all the sectors of science and technology. Application of computers in agricultural research and development has, therefore, become more relevant and important to-day than ever before. Every scientist in the National Agricultural Research System needs to possess the knowledge and skills in working with computers. The Academy offers training programmes in computers and information technology to help achieve this goal. Two programmes on Computer Applications in

Agriculture were organized at the Academy with the aim to expose the participants to the basics of computer and to provide enough hands-on experience to facilitate them develop skills in working with important software packages for use in their back-home situation. The first programme was co-ordinated by Dr K. Vidyasagar Rao and Dr K.M. Reddy, and the second programme was co-ordinated by Dr K.M. Reddy and Dr K. Vidyasagar Rao. A total of 24 scientists and technical officers, from various ICAR institutes and SAUs, were trained through these two programmes.

Computer 17-08-99
Applications 27-08-99
in Libraries
and Infor-
mation
Centres

Manual methods and techniques have become less effective in information management. Consequently, information management in libraries and information centres has become a challenging task to the information managers. The developments in computer-aided information technologies and the advent of user-friendly software for information management have brought about newer dimensions to information communication and management. These breakthroughs in information technology facilitate effective management of information. A course on Computer Applications in Libraries and Information Centres was, therefore, organized, with a view to familiarize the participants with the concepts of DBMS and to develop skills in them in the use of application programs and in designing bibliographic databases. Appropriate blend of lectures, demonstrations, and hands-on-experience was used to cover various aspects that would help in realizing the objectives. Thirteen scientists and officers of different institutions under NARS, who are responsible for managing library and information centres in their institutions, participated in this programme, which was co-ordinated by Dr K. Vidyasagar Rao.

Statistical 05-10-99
Softwares 16-10-99
for Ad-
vanced Data
Analysis

Techniques for advanced analysis involve complex computations and demand considerable time and skill on the part of scientists using huge data and complex analysis procedures. With the advent of user-friendly software packages, the analysis of complex data has become simple and easy. There is, therefore, a need to expose the scientists to the various statistical software packages for data analysis. It is equally important that the results are presented effectively. In order to develop the necessary skills in the scientists to use various statistical packages and to enable them to develop expertise in effective presentation of results, a two-week course on Statistical Softwares for Data Analysis was organized.

		Various topics covered included Experimental design, stability analysis, and non-orthogonal data analysis; Multivariate analysis; Forecasting techniques; Growth analysis; Geographical Information System; Networks, Internet, and Intranet applications; and Non-linear statistical models. Seven scientists and officers from ICAR Institutes and SAUs, involved in data compilation and analysis, attended the programme. Dr K. Vidyasagar Rao and Dr M. Narayana Reddy co-ordinated the programme.
Management Development Programme in Agricultural Research	25-10-99 30-10-99	It was emphasized in different forums that the Academy need to organize Management Development Programmes (MDP) in Agricultural Research for the newly-recruited Heads of Departments and Project Co-ordinators of ICAR Headquarters and various research institutes. Keeping in view the management responsibilities of these research managers, the second MDP in Agricultural Research was organized at the Academy with an aim to provide a broad overview on the various dimensions of research management, human resource management, information management, and financial management. Heads of Departments and Project Co-ordinators, numbering 14, participated in this programme, which was co-ordinated by Dr S.N. Saha and Dr Jagannadham Challa.
Production Systems Approach for Sustainable Agricultural Development	16-11-99 20-11-99	With the production systems approach gaining importance and acceptance over time, it has become necessary to familiarize the scientists and researchers with the basic concept and process of production systems approach for sustainable agricultural development. The course enabled the participants to understand the concept of production systems approach and the state-of-art of its implementation, and also suggested appropriate measures for its effective implementation. Various topics relevant for the proper understanding of production systems approach were covered during the programme. Researchers from different organizations of NARS, numbering six, participated in this programme, which was co-ordinated by Dr T. Balaguru and Dr P. Manikandan.
Improving Administrative and Financial Management in ICAR Institutes	30-11-99 06-12-99	The course enabled the participants to develop an appreciation and understanding of the various issues of administrative and financial management in the ICAR system, with a view to enable them to work positively and in team to achieve the organizational goals.

Administrative Officers and Finance and Accounts Officers of various ICAR institutes, numbering 32, attended the programme. Dr S.N. Saha and Mr B. Pandu Reddy co-ordinated this programme.



Executive Development Programme in Agricultural Research Management 21-12-99
23-12-99

The second Executive Development Programme (EDP) in Agricultural Research Management was organized with a view to provide a forum for the Directors of ICAR institutes to discuss various research management issues confronting the effective management of the institutes. The programme provided an opportunity for the Directors to share their experiences based on individual, specific cases, in addition to the discussion on some practical tools for solving the research management problems. The programme, co-ordinated by Dr S.N. Saha and Dr Jagannadham Challa, had the active participation of nine Directors from various ICAR institutes.

Management of Information Centres 07-02-2000
18-02-2000

Information Centres in organizations are systems developed to meet the need of every individual organization and can participate and function smoothly in global information networks. Information centres focus on managing an organization's information systems and information base. These centres have to be developed to be proactive to fulfil the needs of the information users in an organization. This course provided an opportunity for the people, managing information centres in their organizations, to exchange experience and skills in managing information resources such as hardware, software, people, and communication. Various aspects related to information management were covered through an experiential and interactive process in this course. The course, co-ordinated by Dr Ajit P. Maru, was attended by seven participants from various institutes, who are ARIS Cell in-charges and Network administrators.

Training of Trainers 22-02-2000
04-03-2000

The concept and use of adult learning and experiential learning principles, coupled with the use of advanced training methods, are bound to increase the training effectiveness. This course helped to orient the trainees to the concept and use of the above ingredients in training, with a view to improve their skill and effectiveness. The programme covered various important as

Project Management Using Microsoft Project	21-02-2000 26-02-2000	pects of training planning and management, through an experiential and co-training approach, that facilitated the trainees to see for themselves the effectiveness of these approaches, thereby motivating them to adapt these ideas and methods for use in their back-home situations. The programme, co-ordinated by Dr T. Balaguru and Dr P. Manikandan, had an active participation of five trainers from various institutions. Effective management of research projects is a pre-requisite for efficient utilization of resources and also for effective monitoring and evaluation of projects. With the launching of the National Agricultural Technology Project (NATP), and with the developments in information technology, it has become necessary for the investigators of NATP to learn and use the methods and tools of computer-based project management for effective decision - taking. Keeping this in mind, a training programme on Project Management Using Microsoft Project was organized for the Principal Investigators (PIs) and Co-PIs of NATP. The course covered various aspects related to project management concepts and tools, Microsoft Project tools and techniques, and use of these tools in monitoring resources, budget, and time. The participants were provided adequate opportunity to develop the needed skills through hands-on experience and also through a case study on a large project. Twenty three PIs of NATP were benefited by participation in this course, which was co-ordinated by Dr D. Rama Rao and Dr Ajit P. Maru.
Internet Usage in Agriculture	27-03-2000 01-04-2000	Internet, which has revolutionized our society and contributed to the realization of a global village, is being increasingly used in agricultural research and education. Researchers and workers use the Net not only to source and exchange information but also for a variety of productive purposes. With a view to expose the agricultural scientists to the concept of networks and Internet, and to facilitate them develop adequate skills in using the Internet resources for accessing information related to agriculture, a training programme on Internet Usage in Agriculture was organized. Major aspects which were covered through theoretical orientation and hands-on experience, during the course, included Networks and Internet, Internet addressing, Internet Mail System, Telnet/Remote login, Usenet groups, FTP, WWW, Search engines, Chat, Virtual learning centres, and e-commerce. Ten participants, working in agricultural R & D institutions, participated in this programme. Dr R. Chithraichelvan and Dr Ajit P. Maru co-ordinated this programme.

Sponsored Programmes

Executive	14-04-99
Development	19-04-99
Programme	
on Computer	19-07-99
Applications	23-07-99
in Agricultural	
Research	26-07-99
and Educa-	31-07-99
tion for	
Senior	25-10-99
Faculty of	30-10-99
TNAU	

To sensitize the senior faculty members of the Tamil Nadu Agricultural University (TNAU), Coimbatore, to the basic concepts of computer and its applications, and to develop necessary skills in them in the use of computers for improving the quality of agricultural education, four sponsored, tailor-made programmes

on Computer Applications in Agricultural Research and Education were specifically organized for the senior-level faculty members of TNAU. The participants were provided with ad-



equade hands-on practice with computers to develop skills in Word Processing, Electronic Spreadsheet, Database Management, and Slide preparation and presentation, in addition to exposure on Internet and e-mail. In all, 61 senior faculty members were benefited by participation in these programmes. Dr S.N. Saha, Dr K. Vidyasagar Rao, and Dr R. Chithiraichelvan co-ordinated the first programme. The second programme was co-ordinated by Dr M. Narayana Reddy, Dr Ajit P. Maru, and Dr A. Gopalam. Dr D. Rama Rao and Dr R. Kalpana Sastry were the co-ordinators of the third programme. Dr M. Narayana Reddy and Dr N. Hanumantha Rao co-ordinated the fourth programme.

Executive	24-01-2000
Develop-	30-01-2000
ment	
Programme	28-02-2000
on Compu-	04-03-2000
ter Applica-	
tions in	27-03-2000
Agricultural	01-04-2000
Research	
and Educa-	
tion	

With the advent of information technology, there is a need to expose the administrative and finance personnel for improving their efficiency and effectiveness. Based on the Memorandum of Understanding between NAARM and Tamil Nadu Agricultural University, Coimbatore, three Executive Development Programmes were organized for the administrative and finance personnel of TNAU, on Computer Applications in Agricultural Research and Education. The basic objectives of these programmes were to expose the participants to the basic concepts of computers, and to develop the needed skills in them in various application software packages, including an exposure to e-mail and Internet. The programme provided the needed confidence and working skills in

		the participants to facilitate them use the expertise developed in the back-home situation. The first programme was co-ordinated by Dr K. Vidyasagar Rao, Dr M. Narayana Reddy, and Dr A. Gopalam. Dr M. Narayana Reddy, Dr Ajit P. Maru, and Dr R. Chithirachelvan co-ordinated the second programme. The third programme was co-ordinated by Dr M. Narayana Reddy, Dr N. Hanumantha Rao, and Dr K.M. Reddy. In all, a total of 43 administrative and finance personnel of TNAU were facilitated to develop the needed skills and expertise.
Executive Development Programme in Educational Administration and Management of Financial Issues	24-01-2000	There is a felt need to bring about change in the attitude and behaviour of administrative and finance functionaries. A programme in Educational Administration and Management of Financial Issues was, therefore, developed for the administrative and finance personnel of Tamil Nadu Agricultural University. Various dimensions related to management of agricultural education, economic reforms, financial management, personnel management, personality development, and other general administrative and financial issues formed the major curriculum for this programme, designed with the primary objective of achieving total quality among the administrative and accounts personnel. A total of 71 administrative and accounts personnel of TNAU were facilitated to develop the needed knowledge, skills, and attitude, by their participation in this Executive Development Programme. Three such programmes were organized during this period. Dr K.P.C. Rao, Dr N. Sandhya Shenoy, and Dr A. Gopalam were the co-ordinators of the first programme. The second programme was co-ordinated by Dr K.P.C. Rao, Dr A. Gopalam, and Dr P. Manikandan. Dr K.P.C. Rao, Mr B. Pandu Reddy, and Dr A. Gopalam co-ordinated the third programme.
	30-01-2000	
	28-02-2000	
	09-03-2000	
Faculty Development Programme in Educational Psychology and Behavioural Technology	27-03-2000	Agricultural education perspectives need to be reoriented to meet the changing demands and challenges. There is, therefore, a definite need to bring about attitudinal and behavioural changes among the faculty members imparting agricultural education. They need to be updated with the needed skills to meet the challenges in agricultural education. Keeping this in view, Faculty Development Programmes were organized, specifically for the faculty members of the Tamil Nadu Agricultural University, Coimbatore. Five such programmes were organized during this period under report, as per the Memorandum of Understanding between NAARM and TNAU. A total of 111 faculty members were benefited by these programmes. Dr A. Gopalam, Dr N.
	01-04-2000	
	10-05-99	
	14-05-99	
	26-06-99	
	10-07-99	
	23-08-99	
	06-09-99	
	20-12-99	
	3-01-2000	
	21-02-2000	
	26-02-2000	

Hanumantha Rao, Dr K.P.C. Rao, Dr P. Kumar, Dr P. Manikandan, Dr R. Chithiraichelvan, and Dr R. Kalpana Sastry were associated with the organization and conduct of these different Faculty Development Programmes.

Advanced Extension and Training Methodology for Animal Husbandry Personnel of Andhra Pradesh

02-06-99
12-06-99

This sponsored programme was organized with a view to orient the Animal Husbandry personnel of Andhra Pradesh to the essentials of extension techniques applicable to animal husbandry development, and to identify animal husbandry problems for initiating development programmes. Various aspects related to extension methodologies were covered during the course, with emphasis on PRA techniques specific to animal husbandry.

Twenty personnel from the Department of Animal Husbandry of Andhra Pradesh were benefited by participation in this course. Dr VE. Sabarathnam and Dr S.K. Nanda were the co-ordinators of this programme.

Management of Human Resources

22-11-99
27-11-99
13-12-99
18-12-99

There has been an increasing demand for sensitizing the officers of the Department of Animal Husbandry, Andhra Pradesh, on various issues of human resource management. On the basis of the request received, two advanced courses on Management of Human Resources were organized, specifically for various Assistant Directors of the Department of Animal Husbandry, Andhra Pradesh. These programmes aimed at exposing the participants to the various important dimensions of human resource management and to facilitate them to use the learnings in their back-home situation to manage people more effectively. Emphasis was made in the programme on various practical issues related to different facets of human resource management. Ten Assistant Directors from different districts of Andhra Pradesh participated in the first programme, co-ordinated by Dr P. Manikandan and Dr R. Kalpana Sastry. The second programme, co-ordinated by Dr P. Manikandan and Dr S. Shanmugam, had the participation of 18 Assistant Directors of different districts.



Decision Support Systems in Agriculture	11-10-99 21-10-99	Effectiveness and impact of the decision-making depend on the timely and quality decisions taken. Decision support systems play an important role in effective and efficient planning and management of agricultural research and education. With the tall future challenges to be achieved in agricultural research and educational development, there is a need to sensitize the faculty members, researchers, and scientists of the National Agricultural Research System to the importance and use of Decision Support Systems. With the financial support from ICAR, a short course on Decision Support Systems in Agriculture was, therefore, organized at the Academy. Various aspects related to the understanding and development of skills in Decision Support System were covered during the course. Twenty one scientists and faculty members, from various ICAR institutes and SAUs, participated in this short course. Dr D. Rama Rao and Dr S.K. Nanda were the Course Directors.
Training Course on MS Office and E-mail	06-12-99 10-12-99	This course, specifically planned and organized for the staff members of Directorate of Oilseeds Research, Hyderabad, facilitated the participants to have extensive hands-on experience to develop skills in the use of computers for day-to-day work. Twelve staff members were benefited by participation in the programme. co-ordinated by Dr Ajit P. Maru .
Agricultural Research Management	10-01-2000 15-01-2000	Faculty members and researchers in the National Agricultural Research System need to be exposed to the various dimensions of Agricultural Research Management, that will help them in the better planning and execution of agricultural research. Based on the request received by the Academy from the Chandra Shekar Azad University of Agriculture and Technology (CSAUAT), Kanpur, a special, tailor-made, sponsored programme on Agricultural Research Management was organized for 25 senior Professors and Heads of Departments of CSAUAT, Kanpur. The programme provided a forum for the senior-level Professors to discuss on various issues related to agricultural research management. Different dimensions of this important aspect was covered through a participatory mode of interaction, which benefited the participants to conceptualize various aspects of agricultural research management. The programme was co-ordinated by Dr T. Balaguru and Dr P. Manikandan.
Internet and Intranet Administration	17-01-2000 23-01-2000	A tailor-made, special course on Internet and Intranet Administration was planned and organized for the faculty members and senior officers of Tamil Nadu Agricultural University,

06-03-2000
13-03-2000

Coimbatore. The major objective of this course was to provide an initiation to the use of personal computer and network system in the organization. The programme trained the participants in installation and use of Windows 95, Windows NT4, and Linux 6.1 for providing Intranet and Internet services. During this training, theoretical aspects of all issues such as TCP/IP, DNS, DHCP, Proxy services, Anonymous FTP, Webserver, NNTP server, and Network security were dealt with in great detail. Back office and similar collaborative components of a network were also demonstrated, as installed at NAARM. Two such courses were organized, and a total of 30 participants, comprising Professors, Associate Professors, Librarians, and ARIS Cell in-charges of TNAU, were benefited through participation in these two programmes, co-ordinated by Dr Ajit P. Maru.



Content
Develop-
ment for
Intranets

24-01-2000
31-01-2000
14-03-2000
21-03-2000

In order to facilitate the faculty members and senior officers of TNAU, Coimbatore, develop skills for content development for Intranets, that are to be implemented in TNAU, two specialized programmes on Content Development for Intranets were organized. Various aspects which were covered extensively with necessary theoretical orientation and skill development training included Use of Intranets, Web-based content, Virtual learning and information centres, HTML content using Frontpage Express, Using Frontpage as a personal webserver, and Integrating web sites with e-mail, Ftp, and Netmeeting. In addition, during this training, theoretical aspects of all issues in content development, incorporating multimedia, Website management, Active Server pages and collaborative computing, and Internet security were dealt in great detail. Special attention was paid to the needs of developing and administering Electronic Libraries. Thirty Professors, Associate Professors, Librarians, and ARIS Cell in-charges of TNAU were immensely benefited by their participation in these two programmes, co-ordinated by Dr Ajit P. Maru.

Meetings / Workshops

NATP
Review
Meeting

25-04-99
26-04-99

Under the aegis of National Agricultural Technology Project (NATP), a two-day Review Meeting was organized mainly to review the progress of research programmes under the production systems research component of NATP. A special meeting of the Agro-ecoregion Directors was conducted to review the constraints faced by them during the implementation, and to work out ways and means to overcome them.



Based on the deliberations in the meeting, several useful recommendations emerged for the smooth and effective implementation of various production system-based research programmes under NATP. Besides focusing on issues requiring policy decisions, a major emphasis was made to develop the scientists in good research proposal writing so that the rejection rate of proposals is minimized. As a consequence of this decision, NAARM proposes to train scientists in sound research proposal writing for attracting funding from external agencies. The meeting was attended by the Deputy Directors General, Chairman and Members of various Scientific Advisory Panels (SAPs), Agro-ecoregion Directors and Principal Production System Scientists from the five Agro-ecoregions, Officials from Project Implementation Unit (PIU) of NATP, and scientists from NAARM, NCAP, and IASRI. A total of 80 scientists participated in the meeting. The meeting was co-ordinated by Dr T. Balaguru.

Workshop
on Dynamics of
Labour
Management

15-07-99
18-07-99

One of the most crucial components of human resource management in the agricultural sector is the work force employed in farm and other areas. Management of this unorganized group of labour force poses a number of problems. There is a need to reorient the management approaches to facilitate worker-oriented, empowerment strategies that will help motivating them to improve their efficiency and effectiveness. With a view to apprise the policy makers, scientists, and farm managers on various issues and strategies related to management of labour, a four-day workshop on Dynamics of Labour Management was organized at the Academy. The workshop enabled the participants to discuss various aspects which included characteristics of agricultural

Deans workshop on Management of Agricultural Education	19-08-99 21-08-99	labour, temporary status employees, contractual labour, the nuances of laws, and role of employers and employees in effective management. The workshop was organized by NAARM in collaboration with the Institute of Public Enterprise (IPE), Hyderabad. Twenty six senior-level officials of NARS, who manage farm labour in their respective institutions, participated in this programme. The workshop was co-ordinated by Dr R. Kalpana Sastry and Mr B. Pandu Reddy of NAARM, and Dr P. Jyothi of IPE.
Hindi Workshop	29-11-99 30-11-99	This workshop provided a forum for the Deans to discuss various issues concerning management of agricultural education and to develop strategies for coping with the changes in the agricultural education needs. The workshop had the active participation of 32 Deans. Dr A. Gopalam and Dr S.N. Saha co-ordinated this workshop.
Workshop on Intellectual Property Rights and Agricultural Growth	07-12-99 09-12-99	A Hindi workshop on Rajbhasha Karyanvayan Ke 50 Varsh: Upalabdhiyan Aur Akankshayen (50 years of Official Language Implementation : Achievements and Expectations) was organized by the Academy, which provided a forum for discussing various dimensions of Official Language implementation, the progress achieved so far, the goals yet to be achieved, the hurdles that might come in the way of realizing the goals, and the facilities that are available for effective implementation. A total of 57 participants from various ICAR institutes, associated with Hindi activities in their institutes, took part in the workshop. The workshop was co-ordinated by Dr A. Gopalam.
		With the changes in government policies and economic reforms, there is an increasing demand for sensitizing scientists, policy makers, and others connected with agricultural research on the need, necessity, and processes of patenting and on various other issues related to intellectual property rights. A workshop on Intellectual Property Rights and Agricultural Growth, being the need of the hour, was, therefore, organized at the Academy in collaboration with the Institute of Public Enterprise, Hyderabad. The workshop provided a forum for the participants to discuss about the amendments and enforcement of patent laws, and procedures and processes of patenting, in relation to Indian agriculture. Scientists and decision makers at various levels in the National Agricultural Research System, numbering 25, were benefited through participation and active interaction in this programme. The workshop was co-ordinated by Dr R. Kalpana Sastry and Dr K.P.C. Rao.

Workshop on Pedagogy and Educational Technology

13-12-99
14-12-99

The workshop provided a forum for the faculty members in the NARS to discuss about the various issues and dimensions of Pedagogy and Educational Technology, with a view to facilitate the use of these technologies for improving agricultural education. Prof. Larry E. Miller, Ohio State University, USA, was the facilitator of this workshop. Twenty seven faculty members from various SAUs, including NAARM, participated in this workshop. Dr B.S. Chandel was the co-ordinator of this programme.

Intensive Hindi Training-cum-Workshop

18-01-2000
22-01-2000

This training-cum-workshop was designed with a view to bring about awareness in Official Language Policy of Government of India, and also to impart the needed skills and practical knowledge to the personnel who are not part of the Hindi section. The programme, co-ordinated by Dr A. Gopalam and Ms J. Renuka, was attended by 34 participants.

Interactive Workshop on O & M Reforms for Administrative and Finance Officers of ICAR

13-02-2000
15-02-2000

The workshop on Organizational and Management Reforms for Administrative and Finance Officers of ICAR provided a forum to discuss and develop a thorough understanding of the role of administrative and finance wings for effective agricultural research management in the 21st century, and also to review the different institute management experiences for identifying issues, with a view to bring about improvement and reforms.

Brainstorming Session on Betterment of Institutional Management

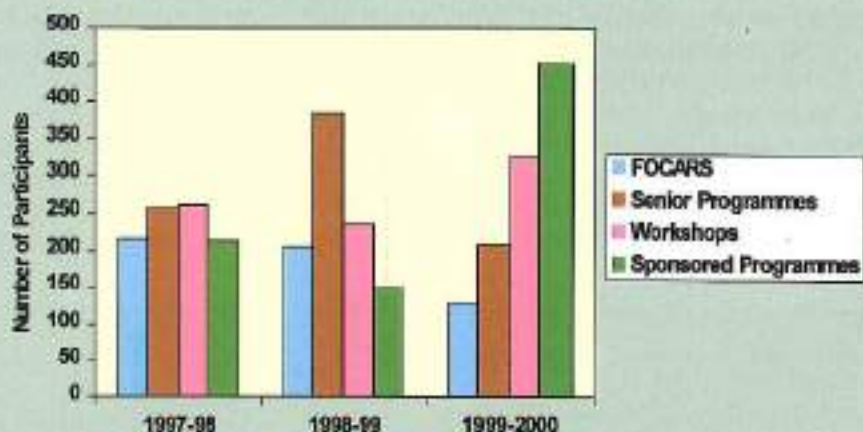
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The interactive workshop was followed by a two-day brainstorming session on Betterment of Institutional Management through Sensitization and Orientation of Administrative and Finance Officers, which facilitated to examine the present structure and systems in ICAR and to identify the changes required to improve institutional management.

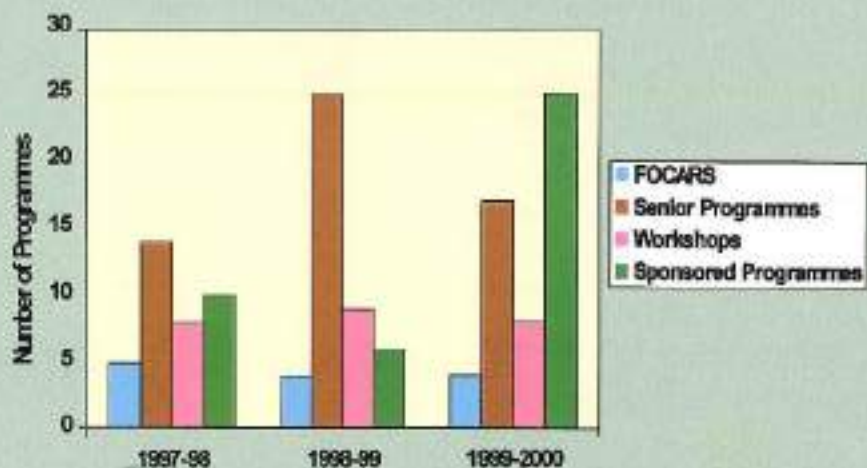


Nearly 150 delegates from ICAR headquarters and from various ICAR Institutes participated in the workshop and brainstorming session. Dr Jagannadham Challa co-ordinated the workshop. The brainstorming session was co-ordinated by Dr K.P.C. Rao and Dr N. Hanumantha Rao.

Participation in programmes during the last three years



Number of programmes organized during the last three years



Feedback

The philosophy of adult learning principles and andragogical process, as applied to training programmes demands that training programmes be evaluated for results to help in reassessing the needs, interests, and values of the participants. The feedback loop is an important mechanism, which makes the training programmes dynamic in nature. The Academy organized over 50 programmes during the period under report. These programmes are classified under four major groups, viz. Foundation Courses, Senior Programmes, Sponsored Programmes, and Meetings and Workshops. Structured as well as oral feedback are collected for every programme, and the constructive suggestions, as brought out both by the written and oral feedback, are used appropriately for future improvement of the programmes. The following table summarizes the major feedback received for different categories of programmes, the strengths being listed under 'Evaluative' and suggestions for improvement being listed as 'Corrective'.

Programme	Feedback	
	Evaluative	Corrective
FOCARS	- Overall content and organization of the programme very good and useful - Skills gained in specific areas for becoming a better researcher	- More time needed for computer practicals - More guest lectures of outside resource persons needed - Visits to more institutes need to be arranged - Need to reduce the number of theoretical lectures - Criteria for FET group formation need a relook
Senior Programmes	- Course material and resource material provided were very useful - Relevance and usefulness of the programmes as well as programme content - Usefulness of exercise sessions and hands-on-practice sessions	- More time needed for practice in courses aimed at skill development and hands-on-experience - Need for more guest lectures from outside resource persons to provide variety - Opportunities to be given to participants for making presentation of their case materials

Sponsored Programmes

- * Relevance of the programme to the needs of the clientele group
- * Course content, modus operandi, and practical utility of the programmes very satisfactory
- * Useful and good study material provided
- * Exemplary behaviour, punctuality, and teaching of faculty members
- * Impressive method of teaching

Necessary prior information to this effect need to be communicated

- * More time needed for library consultation
- * Need for more study visits to institutions in and around

- * Need for more involvement of the trainees in every lecture
- * More synthesis, integration, and summarization of some topics needed
- * More practical examples needed to reiterate the concepts
- * Presentation of some topics needs improvement
- * Need for improving projection facilities
- * Need to provide additional training material like publications, etc.
- * Time duration of different programmes need a re-consideration
- * Resource material of guest speakers are not provided
- * Visit to other institutions less than adequate
- * More case studies and exercises need to be included to ensure variety
- * Refresher courses may be planned for the future

Meetings / Workshops

- * Timeliness and usefulness of the programmes
- * Effective use of case studies in the programmes

- * Structured presentation by participants as case studies need to be included

Research

One of the mandates of the Academy is to undertake systematic review and study of management problems of agricultural research institutes, programmes, and system. In order to fulfil this mandate, the Academy undertakes research in the broad as well as specific areas of agricultural research and education management. The outcome of the research activities facilitate the faculty members to build up high quality resource materials, which add to the credibility of training programmes and workshops conducted by the Academy.

On-going Research Projects

The progress of work in various on-going projects is briefly presented below.

Distance Training for Agricultural Research Management

(Nicole Brunet, ISNAR, and Jagannadham Challa, NAARM)

This project is an ISNAR-ICAR/NAARM research project, in collaboration with Wye College External Programme, UK, and Commonwealth of Learning, Canada. The project is fully funded by Department for International Funding, Govt. of UK, and ISNAR, The Hague, The Netherlands. This joint project was launched on October 1, 1998, and is scheduled to be completed by September 30, 2001.

The higher-level objective of this project is to enhance the contribution of NARS for sustainable development leading to the alleviation of rural poverty in India. The near-term objective is to improve the quality of the Indian NARS to focus agricultural research on poverty alleviation through distance training of researchers and research managers of research stations of SAUs and ICAR institutes. The project is divided into two components, viz. Applied Research and Adaptive Research, and is to finally conclude with an International Conference. The Project Advisory Group, consisting of several Vice Chancellors of SAUs and Directors of ICAR institutes, is headed by Dr S.L. Mehta, Deputy Director General (Education), ICAR.

The project is being implemented in an action-research mode in the following four participating institutions.

1. Acharya N.G. Ranga Agricultural University, Hyderabad, AP
2. University of Agricultural Sciences, Dharwad, Karnataka
3. Mahatma Phule Krishi Vidyapeeth, Rahuri, Maharashtra
4. Directorate of Rice Research, ICAR, Hyderabad, AP

Desk study and field studies were initiated through the following activities:

1. Consultations with NAARM faculty members
2. Design of research tools

3. Initial site visits to the participating institutions for pre-testing of research tools, followed by data collection process
4. Creation of databases
5. Distance Training Strategy workshop in the presence of ISNAR consultant from Wye College, UK
6. Drafting of Applied Research report and printing.

The data analysis of the current context was carried out at several levels, which included

1. Management Training Needs Assessment
2. Review of facilities of information and communication technologies in NARS stations and NAARM
3. Overview of senior level in-service training programmes at NAARM
4. Management training opportunities in India
5. Review of non-formal distance education programmes in India
6. Brief survey on poverty alleviation issues in NARS.

The principal outcomes of the Applied Research component of the project are as follows:

- Three background papers were prepared
- The analysis of the current context was carried out at several levels
- A Distance Training Strategy in agricultural research management for NAARM was designed for implementation in the Adaptive Research Component. The design of Distance Training Strategy included the definition of project vision, mission, and objectives, leading to the selection of a preferred strategy
- The second meeting of the Project Advisory Group was convened on December 14, 1999, where the Applied Research report was presented
- The Applied Research report was printed and published by February 2000

HRD Strategies for Organizational Effectiveness in NARS

(Jagannadham Challa and K. Vidyasagar Rao)

The project, sanctioned under the AP Cess Fund Scheme of ICAR, was initiated from June 1999. The objectives of the project are to develop management systems approach for enhancement of organizational effectiveness through systems of Management by Objectives in five ICAR institutes, including one deemed-to-be-university; to develop and standardize case materials for selected institutions for research productivity and personal development of individuals; and to enhance the co-ordinated effort of research managers and to establish norms for identifying the contribution of individuals.

The data collection is planned through pre-tested research tools and personal interviews, through site visits of the identified institutions. The activities of the project were initiated through extensive literature survey for designing research tools for establishing the current status. The data collection is ongoing for aspects such as HRD climate, competency requirements and training needs, and motivational profiles and needs. Some of the activities, where significant

progress were made, include the following.

- * A base paper on HRD Strategies for NARS was prepared
- * The data collection for HRD climate was initiated through a pre-tested research tool and dispatched to several ICAR institutes
- * The research tool for competency and training needs assessment was designed and it is being planned to collect the data through site visits to five institutes during the coming months
- * A research tool for assessing the motivational profiles and needs is being developed at present
- * A database on the personality attributes of scientists of different cadres and disciplines was completed. The sample size was of 1300 scientists. A similar database is also being developed for Administrative and Finance Officers.
- * About seventy personal interviews with scientists, including research managers, was also completed and compiled.

Studies on Application of Indigenous Technical Knowledge (ITK) in Animal Management for Enhancing Production

(P. Kumar, K. Vidyasagar Rao, and N. Sandhya Shenoy)

The specific objectives of the project are to collect and compile ITK on animal husbandry practices and proverbs, and to classify and document the ITK, species-wise and product-wise. The study envisages compilation of information on indigenous technical knowledge, pertaining to various aspects of animal management, from different parts of the country.

Data have been collected from more than 150 villagers from 14 different states of the country. The collected data are classified theme-wise, viz. practice, beliefs, and disease. The interpretation work is in progress.

Development of National (District level) Agricultural and Natural Resources Database Management System/Management Information System to Assist in Agricultural and Agricultural Research Policy and Management

(N. Hanumantha Rao and M. Narayana Reddy)

Agricultural Management Information System (AGMIS) has been developed using all India district-level data (1991-95) of cereals, pulses, oilseeds, and others (area, production, yield, area under HYV); weather variables; soil parameters, fertilizer use; source-wise irrigation, groundwater (potential and use in 1993), etc. The data can be linked to GIS Arc/Info.

AGMIS was used to develop a water resources and irrigation management strategy for implementing the AP Vision 2020 goals for agriculture in Andhra Pradesh. The recommendations from this study were incorporated in the Andhra Pradesh State Agricultural Policy.

AGMIS is used as a case study and demonstration package for the purpose of training

in Faculty Development Programmes, Management Development Programmes, and in other training programmes of NAARM.

Forecasting Crop Yields Using Time Series Models

(N. Hanumantha Rao)

A software package was developed for forecasting crop yields using Box-Jenkins type of time series models.

Its development and use were demonstrated in senior training programmes of NAARM on Advanced Data Analytical Tools in Agricultural Research and Management.

Expert System for Irrigation Scheduling

(N. Hanumantha Rao, NAARM, and K.V. Rao, CRIDA)

A heuristic expert system for scheduling irrigations to cotton was developed. It uses information about soil characteristics, heuristic information on dryness/wetness of soil, crop appearance, and weather forecasts to provide advice on irrigation decisions.

The expert system was used in NAARM senior training programme on Decision Support System in Agriculture.

Implementation of Farming Systems Research in India : Case Studies

(T. Balaguru and P. Manikandan)

This project was initiated in July 1998, with the following objectives.

- * To investigate the FSR methodology followed by agricultural research institutions in the Indian NARS
- * To identify issues associated with effective implementation of FSR.

In order to fulfil this objective, secondary data pertaining to the implementation of Farming Systems Research (FSR) approach, at the Agricultural Research Station, Aruppukottai, of the Tamil Nadu Agricultural University, were collected. Details on the productivity and profitability of various Integrated Farming Systems (IFS), developed by the Station, have been worked out. The IFS essentially encompasses crops (cotton, sorghum, and pulses), fruits (ber, custard apple, and amla), and livestock (Tellicherry goat) components that are suitable for the dryland region served by the Research Station.

Data collected from the experiments conducted at the Research Station as well as from the trials conducted at the farmers' fields for six years, from 1989 to 1995, are being analysed with particular reference to the extent of incorporation of FSR principle into the methodology followed. In order to identify the constraints faced by the researchers while implementing the programme, it is proposed to visit the Research Station in a couple of months and finalize the Case Study, which could be used in the various training programmes being organized by the

Assessing Impact of Oilseeds Research in India on Production, Prices, and Trade (B.S. Chandel and K.P.C. Rao)

The Project, submitted for funding under AP Cess Fund Scheme, has the following objectives.

- * To evaluate contribution of research on oilseeds production and to estimate returns to investment
- * To identify the impact of research-induced oilseeds production on prices and trade.

A Comparative Study on Performance of GO and NGO KVKs in Agricultural Development with special reference to Andhra Pradesh (N. Sandhya Shenoy and K. Vidyasagar Rao)

The project has been initiated with the following objectives.

- * To study and compare the selected organizational, workgroup, and job involvement variables and roles performed in agricultural development by the GO and NGO KVKs
- * To assess the effect of the selected organizational, workgroup, and job involvement variables on the performance of GO and NGO KVKs
- * To note the expectations of the select clientele groups of the GO and NGO KVKs in view of changing agricultural scenario
- * To find the inter-institutional linkage patterns among the NGO KVKs and other line departments, their expectations, operational constraints, and suggestions for meaningful collaboration

During the period under report, the secondary data were collected on the selected GO and NGO KVKs from the Transfer of Technology Wing of the ICAR, which helped in finalization of the sample. Three different interview schedules were designed, which are for the KVK In-charge, KVK staff, and for the KVK clientele. The designed schedules were modified and standardized with the help of the officials at Zonal Office (Zone - V) of KVK situated at CRIDA. Data were collected from the NGO KVK, Prakasam Krishi Vigyan Kendra, located at Jammikunta, Karimnagar district. The data collected are being tabulated and scored for analysis purpose.

Impact of NAARM Training on ICAR Scientists and Assessment of Their Training Needs (K. Vidyasagar Rao and N. Sandhya Shenoy)

The specific objectives of the project include the following.

- * To evaluate the impact of NAARM training on ICAR scientists and their environment
- * To find out the extent to which NAARM training had helped in job performance of scientists in their work environment

- To assess the benefits accrued by the client institutes from their training alumni of NAARM
- To document the strengths and weaknesses of the NAARM training programmes
- To evaluate the training needs of the ICAR scientists

During the period under report, the data scoring, tabulation, and analysis were undertaken for the FOCARS programme. As the project is of continuous nature, it is decided that the final report will be given after collecting data for two more years.

Research Project Management in the National Agricultural Research System

(P. Manikandan, T. Balaguru, and S.N. Saha)

The project aims at studying the system of research project management in NARS and to suggest a comprehensive system of project management. The project envisages to collect data from selected ICAR institutes, through questionnaire and interview methods, from the in-charges of Technical Cell of the Director, and also from scientists, Heads of Divisions, and Directors. Questionnaires for collecting relevant information were developed for Technical Cell in-charges, scientists, and Heads of Divisions. The data collection is to be initiated.

Concluded Research Projects

Technological Forecasting of Future Oilseeds Scenario in India

(D. Rama Rao and R. Kalpana Sastry)

The project, supported by AP Cess Fund Scheme for a budget of Rs.8.4 lakhs, was initiated on April 1, 1997, and was concluded on March 31, 2000. A brief report of the project and its outcome are presented below.

The objectives of the project are as given below.

- * To forecast oilseeds scenario in the country by 2000 AD and beyond, using technological forecasting methods
- * To analyze various factors/ components/ variables contributing to the oilseeds situation using econometric and systems methods
- * To evaluate comparative performance/ relevance of various forecasting methods in terms of their application in the context of oilseeds in India
- * To provide policy guidelines for strategic planning on oilseeds.

Data on area, production, productivity, HYVs, irrigation, prices, major market centres, and exports and imports of oilseeds were collected from secondary sources. The forecasts were made using a blend of statistical forecasting methods and delphi forecasting.

Project results were discussed with scientists of Directorate of Oilseeds Research, Hyderabad, and presented in a seminar at NCAP, New Delhi, on August 28, 1999. Further, the project results were presented in the National Conference on Oilseeds, organized by the Indian Society for Oilseeds Research, at Hyderabad, on January 22, 2000. The results were presented in the policy session, wherein the issues relating to government policies with regard to R & D, subsidies, processing, prices, trade, knowledge gaps, competing crops, marketing problems, processing, and changing consumer preferences were discussed.

Final report on oilseeds scenario was prepared, after vetting by experts. This includes the past events, present status, and future strategies of almost all the areas associated with oilseeds.

Scope of Forecast

The following entities of oilseeds have been forecasted for 2010 AD: area, yield, consumption, competing crops, constraints of dryland research, influencing sectors, research breakthroughs, limitations on technology transfer, reasons for oilseeds cultivation, seed supply and state role, blending, international trade, processing improvements, and government policies. Some salient findings and strategies, based on the forecasts, are given below.

Production System

The forecasted area under total oilseeds increases from 26 million hectares at present to 32 million hectares by 2010 AD, whereas the average yield increases from 900 kg/ha at present to 1150 kg/ha by 2010 AD. Similarly, the per capita oil consumption (8 kg/annum at present) will be 10.3 kg/annum by 2010 AD.

The forecast for consumption of edible oil in the country by 2010 AD will be 14.6 million tonnes. The oilseed crops will be able to provide 12.3 million tonnes, leaving a gap of 2.3 million tonnes to be met from imports.

The major competing crops for different oilseeds by 2010 AD would be: paddy, sunflower, and millets for groundnut; wheat and millets for sunflower; wheat for rapeseed and mustard; sunflower and gram for safflower; paddy for soybean; groundnut, millets, and pulses for sesame; and millets for castor.

Research Challenges

Slow results of the dryland research, insufficient resources, uncertain rainfall, lack of access to germplasm, lack of appreciation for basic research, and lack of interest in researchers were observed as the major factors affecting dryland oilseeds research.

The time required to evolve oilseed varieties for specific end uses are: for confectionery type in both groundnut and sunflower within 7 years, low erucic acid in rapeseed and mustard in 5 years, spineless variety of safflower in 5 years, bold white seed in sesame in 5 years, and low linoleic acid and double purpose linseed in about 11 and 8 years respectively.

The sub-systems associated with oilseeds that are likely to have more impact on overall oilseed situation by 2010 AD, in the order of importance are: TOT (development), research and education, marketing, and trade and processing.

Achievable R and D Breakthroughs

Some major research breakthroughs that are possible in the next ten years, in each of the oilseeds are: development of HPS variety in groundnut; improvement in seed viability and herbicide application technology in soybean; development of photo insensitive varieties with low senescence and exploitation of germplasm for HYVs development in sesame; improved techniques to augment higher percentage of pure seed and development of alternaria resistant varieties in sunflower; deregulation and liberty in use of TM hybridization technology and breeding alternaria resistant varieties in rapeseed and mustard; identification of cytoplasmic genetic male sterility system to exploit hybrid vigour and standardization of hybrid seed production technology in safflower; good composites and agro-production technologies under marginal lands in niger; early duration hybrids and development of more resistant varieties to lepidopterous pests in castor; and genetic manipulation to make oil edible and techniques for value addition for commercialization in linseed.

Technology Transfer

Low returns, unreliable and poor markets, inadequate and untimely supply of inputs, and inadequate incentives for effective technology transfer were cited as the major limitations for proper transfer of oilseed cultivation technologies to farmers.

Medium farmers are likely to give more preference for oilseed crops cultivation. The reasons for their preference being commercial nature of crop, crop rotation, exposure to improved technologies, and local market demand.

There is a need to increase the seed replacement rate from the present 5-30 per cent to a desired level of 50-60 per cent for different oilseeds. To achieve this, the state agriculture departments have to take a major lead in the supply of seeds to farmers and also priority finance for seed multiplication.

Consumption

The present market share of less than 5 per cent for blended edible oils in total edible oil market is expected to increase to 8 per cent by 2010 AD. The basis for blending of oils, in future, would be health, cost, keeping quality, and taste, in that order.

Among the oils, groundnut, rapeseed and mustard, safflower, niger, castor, and linseed are expected to be cheaper in domestic market by 2010 AD, whereas sunflower, soybean, sesame, and oilpalm will be cheaper in international market.

The possible technical improvements in oilseeds processing, that could be envisaged by 2010 AD, are increased use of solvent extraction and production of pesticide-free byproducts.

The government policies on futures trading, liberalization of seed import, efficient in-land handling and transport, stock decontrol, and development of infrastructural facilities are expected to overcome the constraints faced by processing units in the next ten years.

R and D Strategies and Recommendations for Action

The important issues are abstracted into policy interventions. The evolved policy initiatives are broadly classified into research, extension, seed, and government policies, and the respective recommendations for action are given below.

Policy

* To exploit high yield potential of oilseed crops in summer season, government may encourage special tariffs for electricity and for water under command areas.

(Action: TMO)

* The import duty on edible oils may be stabilized at 30 per cent. This would protect the

interests of the farmers while taking care of domestic demand to protect the interests of consumers. The use or distribution of imported oils or seeds may be restricted to avoid peak harvesting period of oilseeds.

(Action: TMO)

* Poor consumers buy unbranded oil in low volume on daily basis, but at very high price. To meet the nutrition demand of this group, state may encourage supply of blended oils in small pouches of about 20 ml, sufficient for one use.

(Action: TMO)

Seed

* The technology to be followed can be made available to the farmers in the printed form along with seed bags. This would help the farmers to adopt technology for realizing the full potential of a hybrid or high yielding cultivars. Name of seed procurement official must be given in each bag of seed sold and the person be made responsible for its quality. Seed procurement agencies need to be decentralized to bring such accountability. This can be made mandatory for all seed suppliers.

(Action: Seed Corporations)

* Lack of availability of quality seed is one of the major causes of low productivity. Reputation of seed corporations on seed quality is at stake and there is a need to look for new modes of seed production and distribution. Priority finance may be extended to NGOs and young entrepreneurs for seed multiplication and they may be encouraged to work with research institutes as active partners.

(Action: NABARD)

Research

* The basic research efforts in oilseeds have decreased over the years. To meet the international challenges in changing technological environment, strong basic research is essential. To strengthen basic research in the country, at least 50 per cent of resources in ICAR institutes need to be mandated for this, as adaptive research is being carried out in SAUs.

(Action: ICAR)

* Majority of the scientists are much concerned about the time frame for obtaining fruitful results. As dryland research takes a long time to yield fruitful results, the rainfed oilseeds research has been a low priority. Rainfed research needs to be defined and prioritized in each oilseed research institute, both in terms of resource allocation and also in terms of evaluation of scientific output.

(Action: ICAR)

* It is possible for expansion of area under oilseeds in some non-traditional areas. Suitable varieties for such areas are lacking at present. Therefore, location-specific varieties need to be evolved for this purpose.

(Action: SAUs)

* There is a vast scope for value addition in oilseeds, both for domestic use and for export. The R & D projects on value addition in oilseeds (such as recycling of protein-rich, defatted flour) should be given priority funding and other assistance.

(Action: ICAR)

* Special mission-mode programmes need to be initiated to develop specific varieties in rapeseed and spineless varieties in safflower.

(Action: ICAR, under NATP)

Extension

* Farmers need to be educated on post-harvest processes and production technologies. The research and extension sector are not able to use mass media effectively. Private sector need to be encouraged to produce quality material.

(Action: TMO)

* The production and productivity in the low-productive districts in the country should receive special attention by way of implementing all the relevant technologies developed. Technology transfer in such identified pockets may be given to NGOs, on experimental basis.

(Action: TMO)

Over the years, the Academy has gained expertise in various areas of agricultural research and education management. The faculty members of the Academy do take up consultancy projects and services, depending upon the specific needs received from the clientele organizations. Consultancy services of quite varied nature were offered by the faculty members over the last few years. This also includes management support provided to ICAR as and when needed. The consultancy services extended during the last year are briefed below.

Consultancy on Video Production

With the support from UNDP and World Bank, the Academy has established excellent facilities for video production. NAARM has been extending the services of the video production expertise to various institutes in the NARS. The Consultancy Processing Cell of NAARM has recommended costing of Rs.3 lakhs for each outstation production. The project is a package deal involving scripting, shooting, and editing.

During the year under report, the Video Instruction and Production Laboratory (VIPL) of the Academy had undertaken the production of video programmes for Central Plantation Crops Research Institute, Kasargod. One of the productions is on "CPCRI at a Glance", which projects the achievements of the institute in research on crops such as coconut, arecanut, and cocoa. The second production is on "Integrated Management of Diseases and Pests in Coconut", which highlights the pests and diseases management measures developed by CPCRI, with special reference to Root Wilt Disease of coconut. The shooting for both the productions have been completed at Kasargod and also at the Regional Stations of CPCRI at Kayangulam and Vittal.

Earlier, NAARM had completed the production of video programmes for National Research Centre for Soybean, Indore, and for Central Research Institute for Jute and Allied Fibres, Barrackpore.

Management Support to ICAR

At the request of ICAR, management support was provided for improving the effectiveness of ICAR institutes. For this purpose, NAARM developed relevant proforma for collecting information pertaining to the performance assessment at the institute level. These include i) Institute Management Committee (IMC); ii) Research Advisory Committee (RAC); iii) Staff Research Council (SRC); iv) Publication of Annual Report; v) Resource generation; and vi) Fund utilization.

The information collected from the institutes are being analyzed by NAARM for evolving suitable policy-level guidelines. The methodology for this work was developed by Dr T. Balaguru and Dr S.N. Saha.

Support Provided for O & M Reforms

The Academy provides technical and infrastructure support to ICAR in its efforts to bring about O & M reforms. Base papers on selected issues were prepared for the O & M Workshop held at the Academy in February. One of the base papers on Functional Autonomy, prepared by Dr Jagannadham Challa, Dr D. Rama Rao, and Dr Janardan Jee, compares other scientific organizations and points out to the need for bringing about organizational and management reforms in ICAR for realizing full potential of the autonomy. The other base paper on Leading Change in ICAR : O & M Reforms by Dr N. Hanumantha Rao and Dr K.P.C. Rao highlights the need for change in structure, system, and skills.

Centralized Services

NAARM has a number of well-equipped and well-maintained laboratories and various other facilities that help in carrying out the teaching-learning process more effectively. These various facilities have been developed over the years. The specific improvements and developments in some of these facilities, brought about during the period under report, are briefed below.

Computer Section and ARIS Cell

The Computer Section and ARIS Cell of NAARM is a well-equipped facility with a multi-platform network, comprising servers using Sun OS, Solaris, Windows NT 4.0, and LINUX as Operating Systems. This section consolidated on the Optical Fibre Networking in the Academy, spreading it to the Halls of Residence and Scientists Home. In addition, it established collaborative computing using Microsoft Small Business Server and a single interface Intranet for the Academy, as a continuation of its strategy to maintain the Academy's lead in the use of new ICT. Two laboratories were revamped. One is for training lay users who require more supervision in the initial stages of using computers and the other for training in Systems Administration, and Intranet and Internet Administration, Management, and Content Development. The Computer Section's laboratories were used for training over 1000 scientists, faculty members, and officers during the last year.

The NAARM Intranet was re-structured and certain basic approaches such as the user being only two mouse clicks away from any information required and a single web browser-based interface for all applications were introduced. The Intranet now consists of more than 900 web pages, and includes a prototype, fully-functional Global Learning Centre.

One of the main achievements of the Section was in providing training in new areas such as Intranet and Global Learning Centres, and Virtual Centre Development and Management. The Computer Section also supported Y2K compliance, not only within NAARM but also in the NARS, by providing new Y2K compliant packages, especially in Payroll and Farm Wages. A total of 50 copies of Payroll package were supplied to institutions in the ICAR and SAUs.

The Section assisted Central Agricultural Research Institute, Port Blair, in proposing a Local Area Network and setting up of Intranet and Internet connectivity. Many ad-hoc queries from users across the NARS, seeking solutions on various Networking and Operating Systems problems were answered, providing support to the ARIS.

NAARM Health Centre

The NAARM Health Centre, which was formally inaugurated by the Hon'ble Union Minister of State for Agriculture, Shri Hukumdeo Narayan Yadav, on December 18, 1999, is well-equipped to provide outpatient treatment to all the members of the NAARM family, and also to

the staff and their family members of other institutes who are staying inside the NAARM campus. NAARM Health Centre is the only centre at Hyderabad, providing health care to all the retired ICAR employees and their dependent family members settled at Hyderabad and Secunderabad, including regular supply of medicines.

The facilities available in the Health Centre include routine laboratory investigations such as Haemoglobin, Blood Sugar, Blood Urea, S. Creatinine, S. Bilirubin, Blood grouping, ESR, and ECG. All the laboratory investigations are carried out by the Medical Officer with the support of one Pharmacist. Periodical vaccination of staff and their dependents against Hepatitis B and Typhoid was carried out. The Health Centre is being strengthened to expand its services and activities.

Hostel

The Academy has excellent hostel facilities which cater to the board and lodging requirements of senior officials of NARS as well as the delegates and scientist-trainees of various programmes organized by the Academy. Halls of Residence has 150 rooms, comprising 138 single-bed and 12 double-bed rooms. This hostel has the board facilities for 150 people to dine at a time. Additional facilities for entertainment, games, and physical exercises are also housed in this hostel. The Scientists Home has 30 double-bed rooms, comprising 22 AC rooms which include 7 VIP rooms and 8 non-AC rooms. The dining hall of the Scientists Home can accommodate 50 persons at a time for dining.

During the year under report, Halls of Residence catered to the lodging requirements of over 800 persons, whereas nearly 700 persons stayed at different points of time during the year in the Scientists Home. The Hostel facilities of the Academy could generate an amount of Rs.2,81,446/- during the last year.

Various facilities were added to the Hostel during the last year. A new block, containing 60 rooms, was added to the Halls of Residence. The LPG pipeline of the kitchen was changed and two cooking ranges were added to the kitchen. New stainless steel cutlery worth Rs.35,000/- was also added during this period. These helped in the board requirements of a large number of residents of the Hostel, and also in meeting the food arrangements to be made on various special occasions. Three new water coolers along with water purifiers were also added to the hostel facilities during the last year. The LAN connection installed in the Scientists Home and Halls of Residence catered to the computer facilities requirements of scientist trainees during off hours.

Information Centre

The Library and Information Centre provides valuable input related to library and information, in various programmes organized by the Academy. The Information Centre extended the service to the scientist trainees of various training programmes as well as to the other scientists of ICAR and SAUs in the online search through CD-ROMs and other

databases. During the period under report, the Centre was supported with the following facilities to strengthen its input to the various activities of the Academy.

Under NATP, one Pentium-II computer alongwith an inkjet printer was provided to strengthen the infrastructure. In order to enhance the quality of reprographic facility, one Modi Xerox machine with advanced configuration was obtained during the year. This machine could cater to the increased demands of photocopying, needed for training programmes and for other activities. Nearly 1000 reference books related to various aspects of research management, information technology, computers, and other management aspects were added to the already existing collection in the library, in addition to the 500 books of Hindi procured during the year.

Farm Section

The Farm Section consolidated and also expanded its various activities during the period under report. Participative mode of management, with bottom-up approach was practised, which helped in effective management of farm activities.

The Golden Jubilee Park, developed earlier, was maintained effectively. The tree planting activity was continued. The growth parameters of both the tissue-culture and non-tissue culture developed teak plants were recorded, and the growth monitored. A Mound Garden was developed by the side of the Faculty Building. The surrounding parts of this garden were planted with seasonal and perennial flower beds. This garden was inaugurated by the Director on the Independence day. Plants depicting ICAR emblem and Scientists Home, and Halls of Residence and Mera Bharat Mahan were developed in the lawns of the Scientists Home and Halls of Residence respectively. These lawns were planted with various plants. Most of the plant

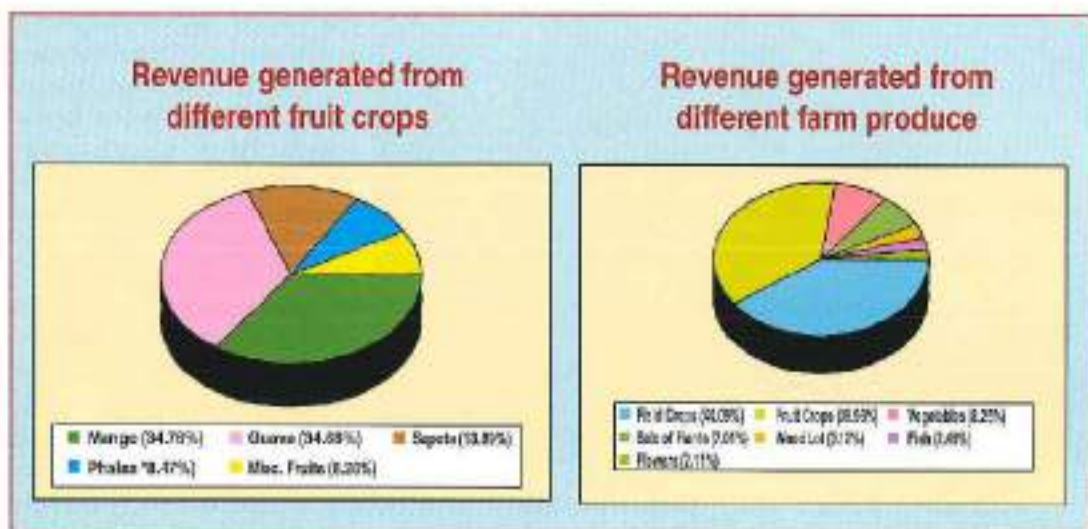


materials used for development were drawn from the multiplied plants of NAARM nursery. Latest model Pop Up sprinklers were fitted in the Administrative Block. Several attractive seasonal and perennial flowers were raised in the lawns of Administrative and Faculty Blocks, which have enhanced the aesthetic look of NAARM campus. At the entrance of the second gate of NAARM, letters 'Welcome to NAARM' were developed in English and Hindi, with various attractive plants.

The entire group of Farm personnel worked as a team and participated in different exhibitions in the twin cities. Several prizes won by the Farm Section, during this period of report, include the following.

- * Second prize under the category of Central Government Institutional Gardens, in the Garden Festival organized by the Department of Horticulture
- * Six first prizes, 8 second prizes, and 5 third prizes, in the XXIV Annual Rose Show conducted by the Hyderabad Rose Society, on December 11 and 12, 1999
- * Best Institutional Rose Garden Trophy, Garden Supervisor Trophy, Trophies for the Best Collection of White, Yellow, Pink, and Lavender roses, in addition to 6 first prizes and 8 second prizes under different sections of flower display, in the XIII Annual Rose Show organized by the Secunderabad Horticultural Society, on December 18 and 19, 1999
- * Second prize under the category of Indoor potted plants, in the Annual Horticultural Show on January 26, 2000
- * Second prize under the category of Rose Cut Flowers (Yellow group) in the X Horticultural Show conducted by the Department of Horticulture, from January 26 to 28, 2000.

The commitment of people and the hard work put forth in the farm activities, by the officers and staff of Farm Section, are reflected in the income generated from the Farm. During the period under report, Farm could generate a total income of Rs.81,853/- from different farm produce.



Other Strength-building Activities

National Agricultural Technology Project

Under the aegis of World Bank - supported National Agricultural Technology Project (NATP), two sub-projects, with a budget outlay of Rs.493.69 lakhs have been sanctioned to NAARM, Hyderabad, for three years from 2000-01.

Sub-project 1 : Institutionalization of Priority Setting, Monitoring, and Evaluation (PME), and Networking of Social Scientists

This is a joint programme, developed by NCAP and IASRI, New Delhi, and NAARM, Hyderabad. The specific components entrusted to NAARM include the following.

(i) **Research Priority Setting at Micro Level**

The major focus of this programme is to develop appropriate research priority setting methods at the micro (institute/production system) level, and to train the scientists in micro-level priority setting.

(ii) **Performance Assessment and Accountability Enhancement of Agricultural Research Organizations in India**

This programme aims at strengthening the NAARM's capacity to deal with issues of institutional performance and accountability through research studies, and institutionalizing improved performance measurement system in the Indian NARS through training.

These two programmes, with a budget outlay of Rs.173.07 lakhs, will be implemented by NAARM in the next three years.

Sub-project 2 : Human Resource Development (HRD) for Agricultural Research and Education Management

The sub-project, with a budget outlay of Rs.320.62 lakhs, will be implemented exclusively by NAARM in the next three years. It is envisaged to undertake four major activities, with a number of specific programmes, as under.

Activity 1 : Training Programmes

Programme 1 : Retreat Programme for Senior Executives of ICAR

This is essentially meant to develop management perspectives and appropriate leadership qualities in the executives towards improved performance of ICAR, in the context of changing global agricultural scenario.

Programme 2 : Management Development Programme (MDP) for Research Managers

The major objectives of this programme is to enhance the managerial effectiveness of research managers in the Indian NARS.

Programme 3 : Training on Research Proposal Writing

The major focus of this programme is to develop the required skills for preparing sound research proposals by assisting the agricultural scientists in translating their ideas into good proposals acceptable to various supporting agencies.

Activity 2 : Research Studies

Programme 4 : Forecasting Trained Agricultural Manpower

Besides providing policy guidelines for planning agricultural education so as to develop needed manpower in different occupations, this programme focuses on training the agricultural scientists/teachers in forecasting manpower needs in agriculture.

Programme 5 : Developing Computer Mediated Courseware for Facilitating Learning - Teaching

This programme aims at developing interactive educational technology learning modules for use in distance education.

Programme 6 : Capacity Building of NARS in Developing Decision Support System (DSS) through GIS

This programme attempts to develop GIS-based DSS for sustainable management of agricultural resources, as well as to use the DSS as training tools for the scientists of NARS.

Programme 7 : Providing Support to PIU of NATP

With a view to providing management support to PIU through identification of day to day implementation problems and offering implementable solutions, this programme is proposed.

Activity 3 : Virtual Agricultural Research and Education Management Campus Development

Through the development of global learning centre in agricultural research and education management, this programme aims at providing opportunities for scientists from India and other country NARS to use educational resources of NAARM in distance education mode using Internet.

Activity 4 : Faculty Development at NAARM

This programme focuses on capacity building of NAARM faculty through training and exposure visits to premier management training institutions in India and abroad.

All the above said programmes, pertaining to the two sub-projects, will be implemented by specifically identified teams of NAARM faculty under the guidance and supervision of the Principal Investigator, Dr J.C. Katyal. In order to facilitate and ensure the smooth functioning of these two sub-projects, Dr T. Balaguru has been nominated as the Nodal Officer of NATP at the Academy.

NAARM Health Centre Inaugurated

The newly-built Health Centre of the Academy was inaugurated by the Hon'ble Union Minister of State for Agriculture, Shri Hukumdeo Narayan Yadav. The Health Centre was inaugurated on December 18, 1999, in the presence of Padma Bhushan Dr R.S. Paroda, Director General, ICAR, and Secretary, DARE. The inauguration of the Health Centre was



followed by felicitation to the Hon'ble Minister by the Directors of all the ICAR institutes located at Hyderabad, by the scientists, and also by distinguished guests. On this occasion, the Hon'ble Minister released the recent publication 'Calendar of Training Programmes', for the year 2000, brought out by the Academy.

While addressing the gathering, the Hon'ble Minister urged the scientists to

introspect the achievements they had made during the past five decades and set their goals afresh to meet the challenges in the new millennium. The Minister urged the scientists to recognize the potential of traditional knowledge which remain unexploited. The Hon'ble Minister emphasized the need to promote scientific outlook among people and also scientific culture in the environment. In the present stage of development, the problems for research are diverse; they represent a wide spectrum, from the simplest to the most challenging and complex ones. There is a stiff fight between the claims made through intellectual property rights and the 'traditional knowledge', and the scientists would have to come up with strategies to harness the power of traditional knowledge, the Minister pointed out.

Dr R.S. Paroda, Director General, ICAR, presided over the function. In his address, Dr Paroda said that it is time to become competitive to take full advantage of the globalization, by self-introspection and full commitment. The Green Revolution made a substantial contribution to food production; but it helped to increase the productivity of only irrigated land. While there is still scope for increasing the yield from irrigated land, we must look at the potential available in waste land and dry land. The scope for ushering in the second Green Revolution exists here only, he pointed out.

Hindi Fortnight Celebrations

Hindi fortnight celebrations were organized at the Academy from September 20 to October 4, 1999, with a view to bring about an awareness about the official language and to create an environment congenial for the use of Hindi for official purposes among the staff and officers of the Academy. The valedictory function was held on October 4, 1999, under the auspices presence of Dr Madugula Nagaphani Sharma, Sahasra- vadhani (both Telugu and Sanskrit), and Chairman of the Official Language Commission of Andhra Pradesh, Hyderabad. While delivering the valedictory address, he explained the importance of language, its aims, and utility. Language is necessary for understanding as well as for study, he asserted. He felt that the day-to-day activities can be performed with the comprehensive use of the existing words. He opined that literacy can bring hope and light in the lives of innocent masses. He advised that everyone should come forward to learn the language voluntarily. While delivering the presidential address, Dr S.N. Saha, Acting Director and Chairman of the Official Language Implementation Committee of the Academy, said that it is not easy to implement the official language (Hindi) in a vast country like India; however, the best efforts are being put forth for its effective implementation. He said that the official language plays a vital role for national integration. Everyone should take part in the effective implementation of official language policy, he emphasized. Earlier, Shri D. Venkateswarlu, Technical Officer (Hindi), welcomed the gathering. Dr A. Gopalam, Officer-in-Charge (Hindi), while presenting the report on the achievements of the Hindi Unit during the year 1998, pointed out that the Academy conducted functional Hindi training to FOCARS trainees and also workshops, which were well received. Several competitions like essay writing, typing, elocution, translation of words, speaking for one minute, memory competition, singing, dictation, noting and drafting, antakshari, etc., in Hindi, were organized. Dr Nagaphani Sharma, the Chief Guest, distributed the prizes to the winners of various competitions. Mrs. J. Renuka, Assistant Director (Hindi), proposed the vote of thanks.



A Creditable Performance by NAARM Sports Contingent

The NAARM sports contingent, which participated in the ICAR Inter-zonal Sports Meet held at ICAR Research Complex, Goa, from May 14 to 17, 1999, secured five individual prizes. In the athletic events (women), the Academy retained the first positions in discus, shot put, and javelin throw. Ms Rukmini Ammal bagged the first prize in shot put and discus throw, and Ms G. Aneeta bagged the first prize in javelin throw. Dr A. Debnath bagged the second

prize in javelin throw (men), and Mr M.K. Samson bagged the third prize in shot put (men). The sports contingent was led by the Sports Manager Dr A. Debnath.

Again, the NAARM sports contingent, which participated in the 16th ICAR (Zone III) Inter-institutional tournaments held at Indian Institute of Horticultural Research, Bangalore, from December 16 to 20, 1999, secured six individual prizes and one team event prize. As in the previous tournaments, this year also NAARM sports contingent came out with flying colours. In the athletic events (women), the Academy retained the first position in shot put. Ms Rukmini Ammal bagged the first prize in shot put, second prize in discus throw, and third prize in javelin throw. In the athletic events (Men), Mr M.K. Samson bagged the second prize in shot put, and third prize in both javelin and discus throws. Also, the NAARM delegation secured the second place in table tennis men's event (doubles). The team comprised Mr Sham Bahadur, Mr S. Swamy, Mr P. Swamy, Mr P.B. Yadaiah, and Mr M. Sekhar Reddy.

Women's Cell

The Women's Cell of NAARM has been performing various activities that help to strengthen the cause of women and also to develop a feeling of cohesiveness and fraternity among the women of the NAARM family. Dr N. Sandhya Shenoy, Senior Scientist, is the President of the Women's Cell.

During the year, the Women's Cell celebrated the International Women's Day by organizing a visit to Shilparamam, a place of cultural importance, for the women employees of NAARM as well as for the Ladies Club members of NAARM campus. The Women's Cell also organized a felicitation to the women employees of NAARM who had shown outstanding performance in sports activities. The cell also looks into the needs and requirements of women employees and also attends to their grievances. The Academy extends its support and co-operation to the cause and activities of the Women's Cell.

Human Resource Development

Dr N. Sandhya Shenoy, Dr B.S. Chandel, and Dr Santosh Kumar Nanda, Senior Scientists of NAARM, were deputed for advanced training to USA under the AHRD Project. The three months training of Dr Sandhya Shenoy was at the University of Illinois, whereas Dr Chandel and Dr Nanda were trained at Cornell University from August 28 to November 28, 1999.

Dr Sandhya Shenoy was trained in the area of Educational Communication, with special reference to Distance Education through Web, at the Agriculture Instruction Media (AIM) Lab at the University of Illinois, Urbana - Champaign, under the guidance of Dr John Schmitz, from May 17 to August 17, 1999. During her stay at Illinois, Dr Sandhya underwent training in various areas which included Research methodology in agricultural education, New tools for Web-based instruction, Distance education, Digital libraries, Virtual classroom, etc. As a part of the training,

Dr Sandhya designed a Web site - The Agricultural Gateway to India - searchable at <http://Web.aces.uiuc.edu/aim/diglib/india>. In addition, a prototype Website for Agricultural Technology Information Centres was also developed, which is searchable at <http://Web.aces.uiuc.edu/aim/diglib/india/atc>.

During the period of his stay at Cornell, Dr Chandel worked under the guidance of Prof. John H. Bishop, School of Industrial and Labour Relations. Dr Chandel worked on various topics which included return to investment in education, performance evaluation and quality management in higher education, evaluation of training, and economic analysis of University. Dr Chandel attended a CIAT International Workshop on "Assessing Impact of Agricultural Research on Poverty Alleviation" at San Jose (Costa Rica) from September 14 to 16, 1999. The travelling allowance to attend the workshop was provided by ISNAR, The Netherlands. A paper titled "Agricultural research and poverty alleviation in India - Management issues" by B.S. Chandel (NAARM) and Paul T. Perrault (ISNAR) was also presented in the workshop.

Dr S.K. Nanda's programme at Cornell was co-ordinated by the NYS College of Agriculture and Life Sciences. Dr David J. Deshler, Professor, Department of Education, was the Faculty Advisor. During the period of training at Cornell, Dr Nanda audited two courses, attended six short-term workshops, participated in three seminars, and also undertook a short project on interdisciplinary research methodology. The results of the research were presented through a paper titled "Emphasis on validation research: meta analysis". The training facilitated Dr Nanda to equip himself with the necessary knowledge, skills, and tools in areas such as teacher development strategy, interdisciplinary research methodology, concept mapping, pattern matching, participatory methods, bibliometric studies, and GIS mapping.

Faculty News

- ✦ Dr J.C. Katyal, Director, NAARM, proceeded on extraordinary leave for one year to work as Senior Research Fellow at the Centre for Development Research, University of Bonn, Germany. Dr Katyal was relieved on June 4, 1999, to take up this consultancy. Dr S.N. Saha, Joint Director, NAARM, was the Acting Director during this period of assignment of Dr Katyal.
- ✦ Dr S.N. Saha, Joint Director, NAARM, participated in the Workshop on Financial Management Systems for Indian Council of Agricultural Research, at National Institute of Financial Management (NIFM), Faridabad, on August 12 and 13, 1999.
- ✦ Dr S.N. Saha, Joint Director, NAARM, has been nominated by Director General, ICAR, as a Member of the Management Committee of Directorate of Rice Research, Hyderabad, for a period of three years from August 2, 1999.
- ✦ Dr T. Balaguru, Principal Scientist, participated, as a consultant, in the National Workshop on Biovillages : A Human Centred Approach to Sustainable Agriculture and Rural Development, organized by M.S. Swaminathan Research Foundation, at Pondicherry, from November 29 to December 1, 1999. On the basis of the recommendations, Dr Balaguru assisted the Foundation in preparing the Manual on Biovillages.

- ✦ Dr P. Manikandan, Principal Scientist, has been nominated by Director General, ICAR, as a Member of the Management Committee of Directorate of Oilseeds Research, Hyderabad, for a period of three years from September 17, 1998.
- ✦ Dr P. Manikandan, Principal Scientist, was awarded Postgraduate Diploma in Human Resource Management (PGDHRM), by the Indira Ganadhi National Open University (IGNOU), New Delhi.
- ✦ Dr K.P.C. Rao, Principal Scientist, participated in the National Seminar on Oilseeds and Oils, organized by the Indian Society of Oilseeds Research, held from February 2 - 4, 2000, at Hyderabad.
- ✦ Dr K.P.C. Rao, Principal Scientist, participated in the Workshop on Research Evaluation and Impact Assessment, conducted by ICRISAT, Hyderabad, from February 10 -12, 2000.
- ✦ Dr K.P.C. Rao, Principal Scientist, visited the ICAR Research Complex for Goa, at the request received from the institute, and trained the scientists in preparing research budgets based on Project-based Budgeting System.
- ✦ Dr M.M. Anwer, Principal Scientist, was relieved on June 30, 1999, to join as Director (Distance Education), Moulana Azad National Urdu University, Hyderabad. Dr Anwer was granted lien for one year to take up this assignment.
- ✦ Dr M.M. Anwer, Principal Scientist, was awarded Postgraduate Diploma in Management (PGDIM), by IGNOU, New Delhi.
- ✦ Dr D. Rama Rao, Principal Scientist, attended the Project Advisory Committee Meeting of the DST Project Assessment of Brain Drain Among the JNTU Colleges of Engineering Graduates - Methodologies and Monitoring, held at JNTU, Hyderabad, on July 16 and on October 22, 1999. He also attended the Project Advisory Committee Meeting of the Project Quantification of Financial and Manpower Resources Devoted to R & D in Science and Technology from Academic Sector South Zone, held at Tamil University, Tanjore, on August 28, 1999.
- ✦ Dr M. Narayana Reddy, Principal Scientist, acted as an external expert in the selection committee for selecting Research Fellows and Research Associates at CRIDA, Hyderabad.
- ✦ Dr Janardan Jee, Principal Scientist, participated in the World Potato Congress held at New Delhi, from December 6-11, 1999.
- ✦ Dr R. Chithiraichelvan, Principal Scientist, underwent a training in Maintenance of Sony Betacam Video Equipment, at M/s Telerad, Ahmedabad, from September 6 - 9, 1999.

- ✦ Dr R. Kalpana Sastry, Senior Scientist, attended a technical workshop on Labour Laws, at Centre for Research Planning and Action (CERPA), New Delhi, from June 9 to 11, 1999.
- ✦ Dr R. Kalpana Sastry, Senior Scientist, participated in the one-day Workshop on Procurement Guidelines for NATP, held at New Delhi, on August 28, 1999.
- ✦ Dr R. Kalpana Sastry, Senior Scientist, successfully completed a distance learning course on Intellectual Property Rights, offered by World Intellectual Property Organization, Geneva.
- ✦ Dr B.S. Chandel, Senior Scientist, attended a Workshop on Impact Assessment of Agricultural R & D, organized jointly by NCAP and ICRI SAT, at ICRI SAT, Hyderabad, on February 11 and 12, 2000.
- ✦ Dr B.S. Chandel, Senior Scientist, has been awarded Diploma in Management, by IGNOU, New Delhi.
- ✦ Dr Ajit P. Maru, Senior Scientist, attended a meeting organized for developing long-term strategies for distance education at IRRI, Philippines, from April 14 to 18, 1999.

Human and Financial Resources

Human Resources

Category	Sanctioned Strength	Posts Filled (as on 31-3-2000)	Vacant Positions
Scientific	43*	24	19
Technical	86	61	05
Administrative	48	47	01
Supporting	41	41	00

*Includes three RMP positions

Subject-wise and post-wise approved categorization for the scientific posts is given below.

Discipline	Cadre Strength			
	Scientist	Senior Scientist	Principal Scientist	Total
Agricultural Economics	-	2	1	3
Agricultural Extension	-	3	1	4
Agricultural Research Mgmt*	-	4	4	8
Agricultural Statistics	-	1	1	2
Computer Applications	-	2	2	4
Educational Psychology*	-	1	1	2
Educational Technology*	-	3	2	5
Human Resources Management*	-	4	1	5
Mass Communication*	-	1	-	1
Research Project Management	-	1	4	5
Rural Sociology*	-	-	1	1
Total	-	22	18	40

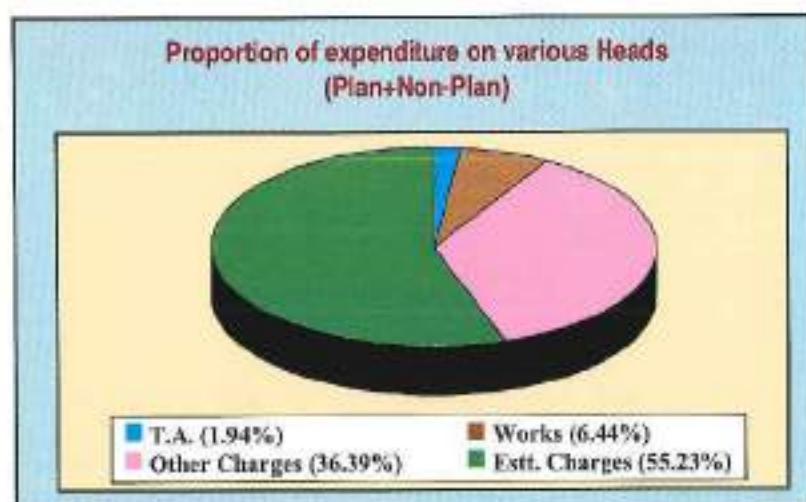
*Discipline does not comply with the Disciplines of Agricultural Research Service

Financial Resources

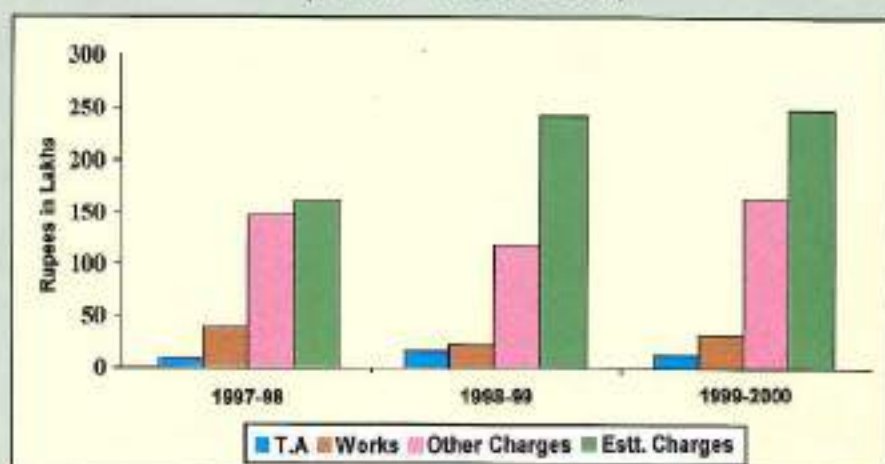
The budget allocation and expenditure during 1999-2000 are given in the following table. Information presented in the table below, also provides details on finance available from other sources.

(Rs. in lakhs)

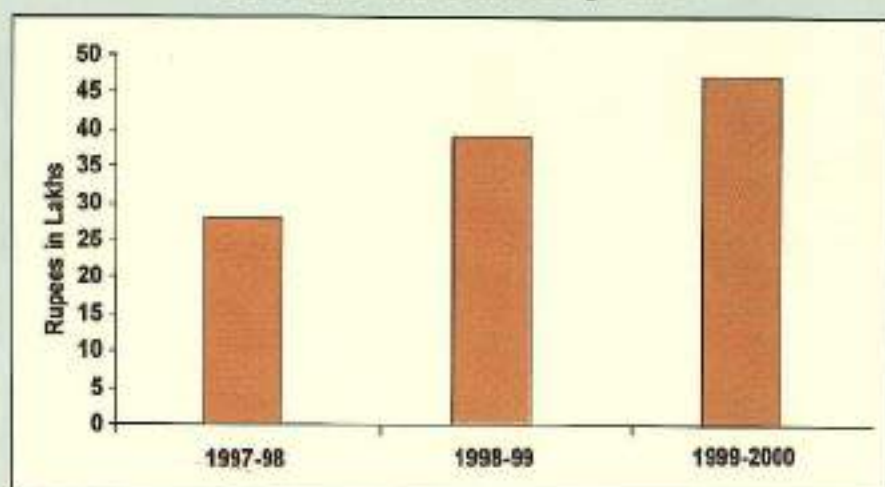
Scheme	Budget allocation	Expenditure
Non-plan		
□ Establishment Charges	219.10	218.87
□ Travelling Allowances	2.70	2.67
□ Other Charges	83.20	82.24
□ Works	—	—
Sub-Total	305.00	303.78
Plan		
□ Establishment Charges	25.00	24.95
□ Travelling Allowances	6.00	5.88
□ Other Charges	78.50	78.41
□ Works	50.50	26.45
Sub-Total	160.00	137.69
Other Sources		
□ A.P.Cess	11.95	5.15
□ AHRD	44.73	33.81
□ NATP	41.71	7.42
Sub-Total	98.39	46.38
Grand Total	563.39	487.85



Proportion of expenditure on various Heads over the last three years (Plan + Non-Plan)



Resource generation over the last three years



Committees

Management Committee

The Academy is guided by the Management Committee in planning, organizing, and conducting its various programmes. The Management Committee approves various programmes and financial outlays thereupon. The Academy ensures their implementation, both in terms of physical targets and schedules, within the sanctioned budget heads. The Management Committee also supports and guides in formulation and consideration of proposals for Five Year Plan and Annual Plans, and takes up periodical review of progress of development schemes. During the year under report, two meetings of the Management Committee were held, the first one on July 31, 1999, and the second meeting on January 22, 2000.

The following is the composition of the Management Committee of NAARM.

Chairman

Dr Saha, S.N., Acting Director, NAARM

Members

Dr Anupam Verma, Dean, Indian Agricultural Research Institute, New Delhi
Mr Jannath Hussain, Secretary, Dept. of Agriculture, Govt. of A.P., Hyderabad
Dr Jha, D.N., Director, NCAP, IASRI Campus, New Delhi
Prof Joshi, B.D., R.K.College, Madhubani, Bihar
Dr Kannalyan, S., Dean, Postgraduate Studies, TNAU, Coimbatore
Dr Krishnaiah, K., Director, DRR, Hyderabad
Dr Mehta, S.L., Deputy Director General (Education), ICAR, New Delhi
Dr Mruthyunjaya, Assistant Director General, ICAR, Krishi Bhavan, New Delhi
Mr Pandu Reddy, B., F & A.O., NAARM, Rajendranagar, Hyderabad
Ms Sabita Das, Ex-MP, Meharpur (P.O), Silchar District, Kachar, Assam
Dr Subba Rao, L.V., Vice-Chancellor, Acharya N.G.Ranga Agricultural University, Hyderabad
Mr Janakiramaiah, K., Asst. Administrative Officer, NAARM
Mr Shankar Rao, Y., Asst. Administrative Officer, NAARM - Member Secretary

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and the rules and regulations of the Government of India that are issued from time to time. The Committee met four times during the period under report, which were held on April 5, July 6, October 13, 1999, and on January 31, 2000. The Committee discussed several issues relating to the implementation of Hindi language during these meetings. The following is the constitution of the Official Language Implementation Committee.

Patron

Dr Katyal, J.C. , Director, NAARM

Chairman

Dr Saha, S.N. , Joint Director

Vice-Chairman

Dr Gopalam, A. , Principal Scientist

Members

Dr Anwer, M.M., Principal Scientist (Up to June 30,1999)

Dr Chandel, B.S. , Senior Scientist

Ms Dhana Laxmi, D. , Library Assistant

Dr Jagannadham Challa, Principal Scientist

Mr Janakiramalah, K., Assistant Administrative Officer

Dr Maru, Ajit P., Senior Scientist

Mr Pandu Reddy, B. , Finance & Accounts Officer

Dr Shanmugam, S. , Principal Scientist

Ms Vanisri, T. , Stenographer

Mr Pradeep Singh, S. , Asst. Director (OL), Co-opted Member

Ms Renuka, J. , Asst. Director (OL), Co-opted Member

Mr Venkateswarlu, D. , Technical Officer (Hindi) - Member Secretary

Visitors

Dr J.R.V. Daane, Director, International Centre for Development Oriented Research in Agriculture, Wageningen, The Netherlands, visited the Academy on May 27, 1999, to explore the possibilities of future collaboration with NAARM and ICAR, in the area of human resource development.

A batch of 20 farmers along with two officers from Devanagiri, Karnataka State, visited the Academy on July 27, 1999, to know the extension activities of the Academy.

Mr Bhagwan Shankar, Special Secretary, Govt. of Uttar Pradesh, Department of Agriculture, Lucknow, visited the Academy on September 23, 1999.

Eight-member Bangladesh delegation visited the Academy on October 4, 1999. The delegation consisted of senior-level agricultural scientists and extension personnel. They interacted with the Director and faculty members on management of agricultural research and extension in India.

A six-member delegation of Board of Regents from University of Agricultural Sciences, Bangalore, visited the Academy on October 21, 1999.

A high-level External Review Team of ICRISAT, under the Chairmanship of Dr H.K. Jain, visited the Academy on November 9, 1999. The team had detailed discussion with the Director and faculty members on many areas of agricultural research management, with particular emphasis on information technology, public awareness, training, and library services.

A batch of Master Trainers from KVK/FTC and State Departments of Agriculture visited the Video Production Laboratory at the Academy on November 18, 1999.

A total of 548 students visited the Academy during this period. These students belonged to various colleges and campuses of different universities which include Tamil Nadu Agricultural University, Tamil Nadu Veterinary and Animal Science University, Pandit Jawaharlal Nehru College of Agriculture and Research Institute, Karalkal, Kerala Agricultural University, Sri Venkateswara University, Tirupathi, University of Agricultural Sciences, Bangalore, Punjab Agricultural University, and Mahatma Phule Krishi Vidyapeet.

Two Iranian scientists, Mr Abdul Rehaman Chiraghali and Mr Mohammad Hadi Khazaie, visited the Academy on March 8, 2000, and discussed with the Director and faculty members on various aspects related to agricultural research management.

Research / Review / Conference Papers

Nanda, S.K., Rivas, A.L., Trochim, W.M. and Deshler, J.D. 2000. Emphasis on validation in research: A meta analysis. *Scientometrics*, 48, 45-64.

Sabarathnam, V.E. 1999. Role of extension in projects oriented to development of technology. *Agric. Extn. Review*, 11, 10-20.

Vijaya Kumari, R. and Rao, K.P.C. 1999. Ex-ante economic evaluation of randomly planted trees in watershed development project areas. *Agric. Econ. Res. Review*, 12, 36 - 47.

Vijaya Kumari, R., Rao, K.P.C. and Chandra Reddy, T. 1999. Distribution of farmers according to net farm returns - A comparative study between watershed and non-watershed areas. *Asian Economic Review*, 41, 172 -177.

Alagarswamy, G. and Narayana Reddy, M. 1999. Simulation modelling and Geographic Information System - New tools in dryland agricultural research. In : *Fifty Years of Dryland Agricultural Research in India*. Singh, H.P., Ramakrishna, Y.S., Sharma, K.L., and Venkateswarlu, B. (eds.). Central Research Institute for Dryland Agriculture, Hyderabad, 249 -258.

Basappa, H., Kalpana Sastry, R. and Harvir Singh. 1999. Cultural practices in integrated pest management of sunflower. In : *Integrated Pest Management in Sunflower*. Basappa, H., Harvir Singh, and Chattopadhyay, C. (eds.). Directorate of Oilseeds Research, Hyderabad, 48-52.

Katyal, J.C., Sharma, K.L., Narayana Reddy, M. and Neelaveni, K. 1999. Integrated nutrient management strategies for dryland farming system. In : *Fifty Years of Dryland Agricultural Research in India*. Singh, H.P., Ramakrishna, Y.S., Sharma, K.L. and Venkateswarlu, B. (eds.). Central Research Institute for Dryland Agriculture, Hyderabad, 357-368.

Rao, K.P.C. 1999. Development of dryland agriculture : Policy issues. In : *Fifty Years of Dryland Agricultural Research in India*. Singh, H.P., Ramakrishna, Y.S., Sharma, K.L. and Venkateswarlu, B. (eds.). Central Research Institute for Dryland Agriculture, Hyderabad, 565-572.

Jagannadham Challa, Rama Rao, D. and Janardan Jee. 1999. Functioning and operational autonomy of selected scientific institutions in India a brief survey. Paper presented at the Second Meeting of the sub-committee on Administrative Matters under the aegis of Organizational and Management reforms task force, ICAR, New Delhi, on October 12, 1999.

Chandala, V.S. and Janardan Jee. 1999. Management of potato pest using bio-control agents. Paper presented at Global Conference on potato, held at New Delhi, from December 6 to 10, 1999.

Chandel, B.S. and Perrault, P.T. 1999. Agricultural research and poverty alleviation in India - Management issues. Paper presented in CIAT International Workshop on Assessing Impact of Agricultural Research on Poverty, held at San Jose (Costa Rica), from September 14-16, 1999.

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Resource Papers

- * Why and How of Training (Balaguru, T.)
- * Team Teaching in Agriculture (Balaguru, T.)
- * Research Publications and Citation as a Measure of Research Productivity (Chandel, B.S.)
- * Economics of Agricultural Education (Chandel, B.S.)
- * Self-directed Performance - Improvement Indicators for SAUs (Chandel, B.S.)
- * Basics of Computers (Chithiraichelvan, R.)
- * Computer Applications in Agricultural Research (Chithiraichelvan, R.)
- * Information Access Through Internet (Chithiraichelvan, R.)
- * Embedding the Sustainability Perspective into Agricultural Research Management (Hanumantha Rao, N., Katyal, J.C. and Narayana Reddy, M.)
- * Operationalizing Values and Value Systems in NARS (Hanumantha Rao, N.)
- * Identifying the Opportunities Provided by e-commerce to Agribusiness and Research (Hanumantha Rao, N.)
- * Effective Farm Management - A Viable Proposition? - A Case Study (Kalpana Sastry, R. and Manikandan, P.)
- * Agricultural Scientists Perceptions on Indian Agriculture : Scene, Scenario, and Vision (Katyal, J.C., Narayana Reddy, M. and Hanumantha Rao, N.)
- * Systems Approach to Training (Manikandan, P.)
- * HRM Perspectives for AOs and FAOs (Manikandan, P.)

- Personality Development (Manikandan, P.)
- Applications of GIS in Agricultural Research (Narayana Reddy, M.)
- Office Automation (Narayana Reddy, M.)
- Desktop Information Management with Exercises (Narayana Reddy, M.)
- Linear Optimization Using MS Excel with Exercises (Narayana Reddy, M.)
- Exercises on General, Graphical, and Statistical Features of MS Excel (Narayana Reddy, M.)
- Agricultural Research Evaluation Methods (Rao, K.P.C.)
- Integrating Agri-business Management into the Curriculum of Agricultural Education (Rao, K.P.C.)
- Vigilance Procedures (Sabarathnam, V.E.)
- Communication Skills (Sabarathnam, V.E.)
- Guidelines for Field Experience Training (Sabarathnam, V.E.)

Training Resource Materials

- Thirty eight training videos were produced by VIPL, involving the scientist-trainees of various FOCARS batches.
- Five slide-tape modules were prepared, involving the scientist-trainees of various FOCARS batches. These are i) Cut Flowers - Increasing Vase Life; ii) Oyster Mushroom Cultivation; iii) Environmental Pollution Due to Intensive Agriculture; iv) Bio-diversity for Development; and v) Insecticide Impact on Agriculture and Society.

The technical back-stop for both these ventures of video tapes and slide-tape modules were provided by the concerned faculty members.

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T.V. Ramdas, Stenographer

Joint Director's Office

S.N. Saha, Joint Director (Acting Director from June 5, 1999)

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T. Balaguru, Principal Scientist

N. Sandhya Shenoy, Senior Scientist

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Janardan Jee, Principal Scientist

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N. Hanumantha Rao, Principal Scientist

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